MATH 244: final project

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.6
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.1     ✔ tibble    3.3.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(infer)
library(ggplot2)
library(dplyr)
library(GGally)
```

```
wildfires <- read_csv("/Users/cristiancastellanos/Downloads/wildfires_2025.csv")
```

```
New names:
Rows: 19210 Columns: 20
── Column specification
──────────────────────────────────────────────────────────── Delimiter: "," chr
(1): Wildfire dbl (18): ...1, latitude, longitude, pr, rmax, rmin, sph, srad,
tmmn, tmmx,... date (1): datetime
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this message.
• `` -> `...1`
```

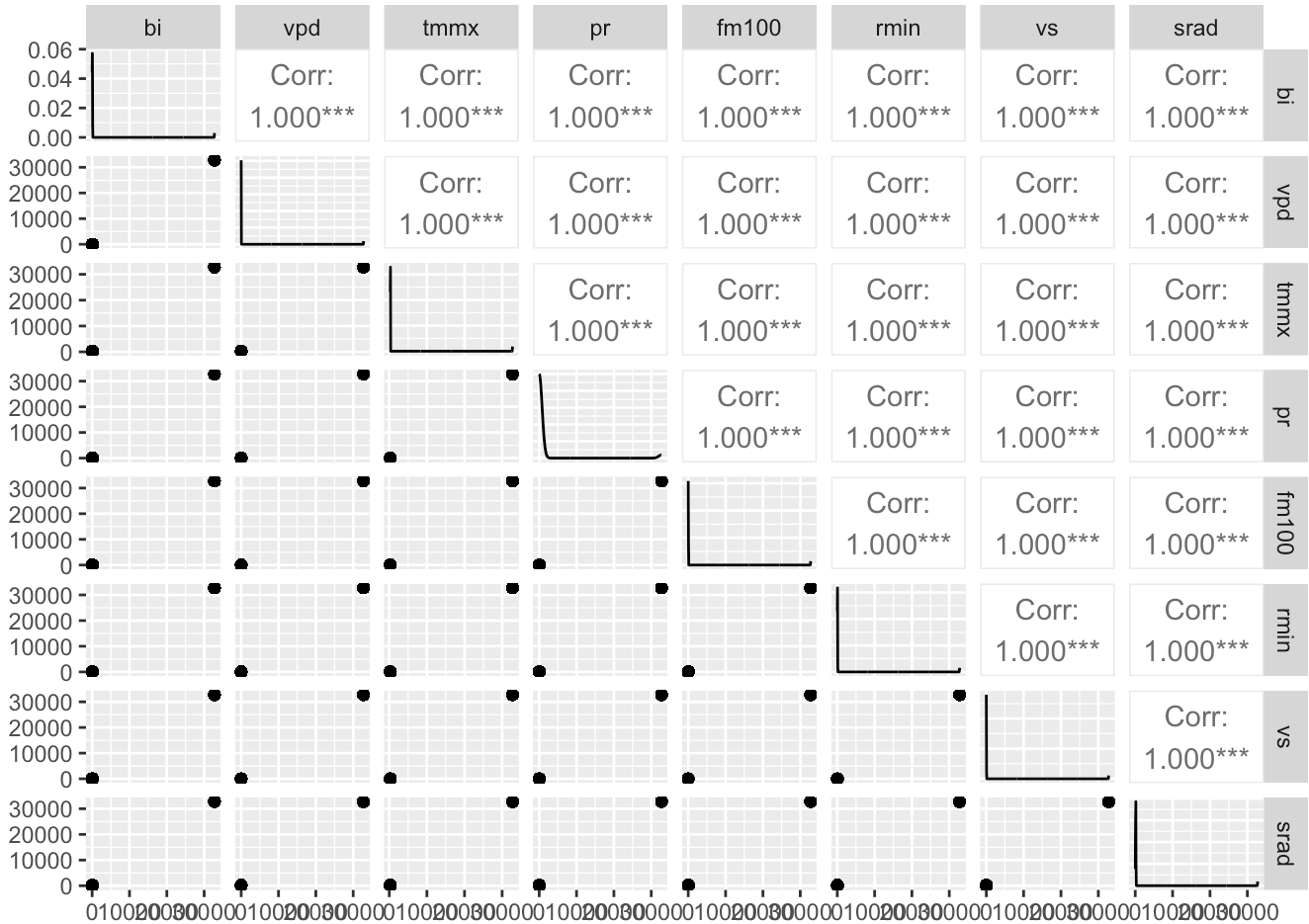
```
dim(wildfires)
```

```
[1] 19210    20
```

```r
# 19210 x 20
colnames(wildfires)
```

```
 [1] "...1"      "latitude"  "longitude" "datetime"  "Wildfire"  "pr"
 [7] "rmax"      "rmin"      "sph"       "srad"      "tmmn"      "tmmx"
[13] "vs"        "bi"        "fm100"     "fm1000"    "erc"       "etr"
[19] "pet"       "vpd"
```

```r
wildfires %>%
  select(bi, vpd, tmmx, pr, fm100, rmin, vs, srad) %>%
  ggpairs()
```




```
str(wildfires) # -> check data types, all values are there
```

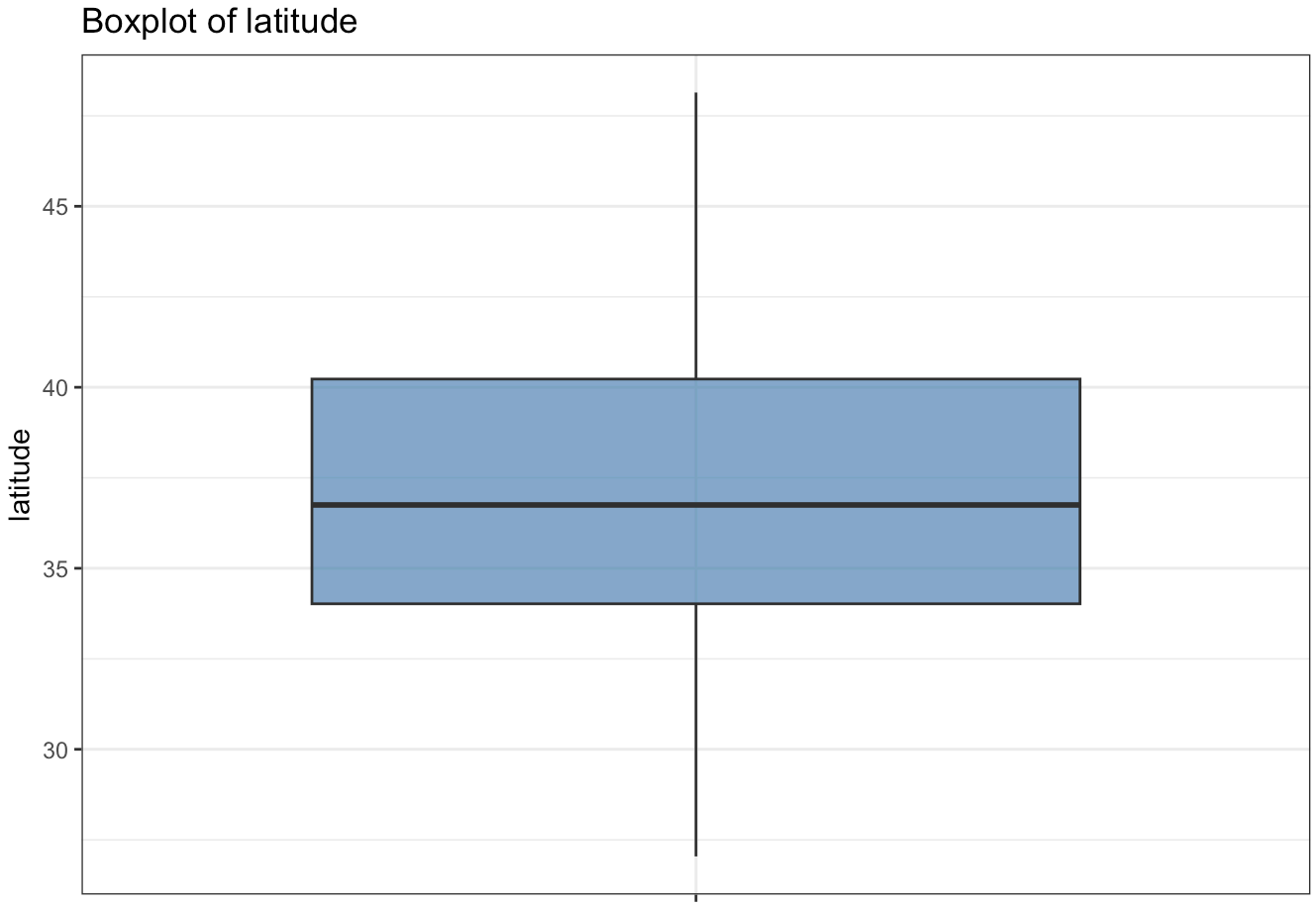
```
spc_tbl_ [19,210 × 20] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ ...1     : num [1:19210] 7246 7247 7248 7249 7250 ...
 $ latitude : num [1:19210] 33.6 33.6 33.6 33.6 33.6 ...
 $ longitude: num [1:19210] -116 -116 -116 -116 -116 ...
 $ datetime : Date[1:19210], format: "2025-01-01" "2025-01-02" ...
 $ Wildfire : chr [1:19210] "No" "No" "No" "No" ...
 $ pr       : num [1:19210] 0 0 0 0 0 0 0 0 0 0 ...
 $ rmax     : num [1:19210] 45.8 42.1 26.7 51.7 37.1 47.4 37.6 37.3 29.6 33.1 ...
 $ rmin     : num [1:19210] 9 7 2.3 10.3 5.8 9.7 13.5 10.3 3.7 4.9 ...
```

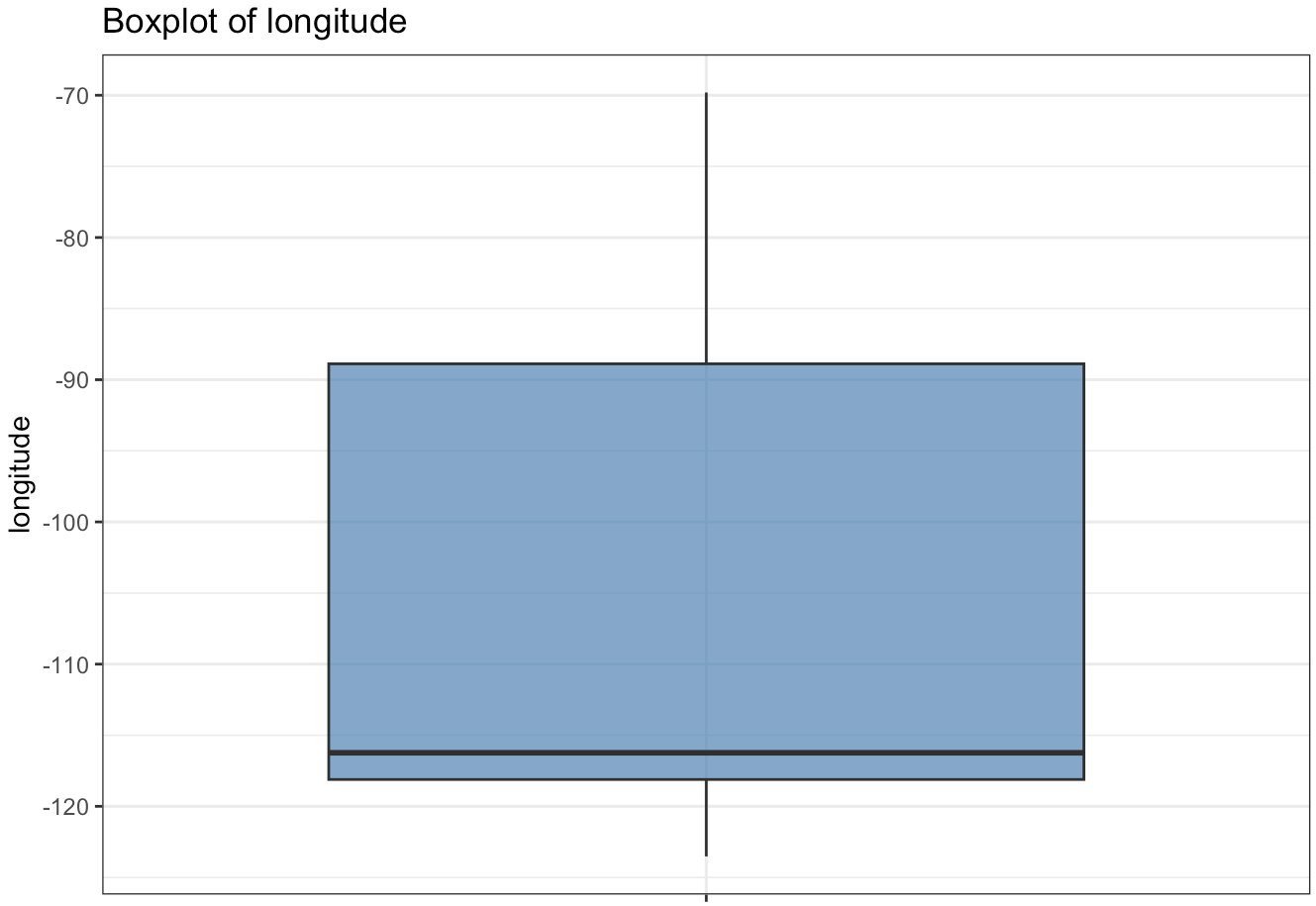
```
 $ sph      : num [1:19210] 0.00248 0.00213 0.00139 0.00284 0.00205 0.00263 0.00278 0.00251
0.00162 0.00162 ...
 $ srad     : num [1:19210] 141 142 129 142 129 ...
 $ tmmn     : num [1:19210] 278 277 278 278 279 ...
 $ tmmx     : num [1:19210] 297 297 298 298 298 ...
 $ vs       : num [1:19210] 2.2 2.1 2.3 3.5 3 2.6 6 7.1 5.1 2.9 ...
 $ bi       : num [1:19210] 53 54 60 67 66 61 87 100 89 68 ...
 $ fm100    : num [1:19210] 9.1 8 6.6 6.8 6.2 6.3 6.1 6 5.4 5.1 ...
 $ fm1000   : num [1:19210] 7.8 7.8 7.6 7.5 7.3 7.3 7.2 7.1 7 6.9 ...
 $ erc      : num [1:19210] 75 78 83 81 83 83 81 84 87 88 ...
 $ etr      : num [1:19210] 4.7 4.7 5.4 6.6 6.4 5.3 8.4 9.8 9.2 5.8 ...
 $ pet      : num [1:19210] 2.9 2.8 3.3 4 3.8 3.3 5.1 5.9 5.5 3.5 ...
 $ vpd      : num [1:19210] 1.53 1.55 1.78 1.55 1.7 1.52 1.43 1.53 1.73 1.5 ...
 - attr(*, "spec")=
  .. cols(
  ..   ...1 = col_double(),
  ..   latitude = col_double(),
  ..   longitude = col_double(),
  ..   datetime = col_date(format = ""),
  ..   Wildfire = col_character(),
  ..   pr = col_double(),
  ..   rmax = col_double(),
  ..   rmin = col_double(),
  ..   sph = col_double(),
  ..   srad = col_double(),
  ..   tmmn = col_double(),
  ..   tmmx = col_double(),
  ..   vs = col_double(),
  ..   bi = col_double(),
  ..   fm100 = col_double(),
  ..   fm1000 = col_double(),
  ..   erc = col_double(),
  ..   etr = col_double(),
  ..   pet = col_double(),
  ..   vpd = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

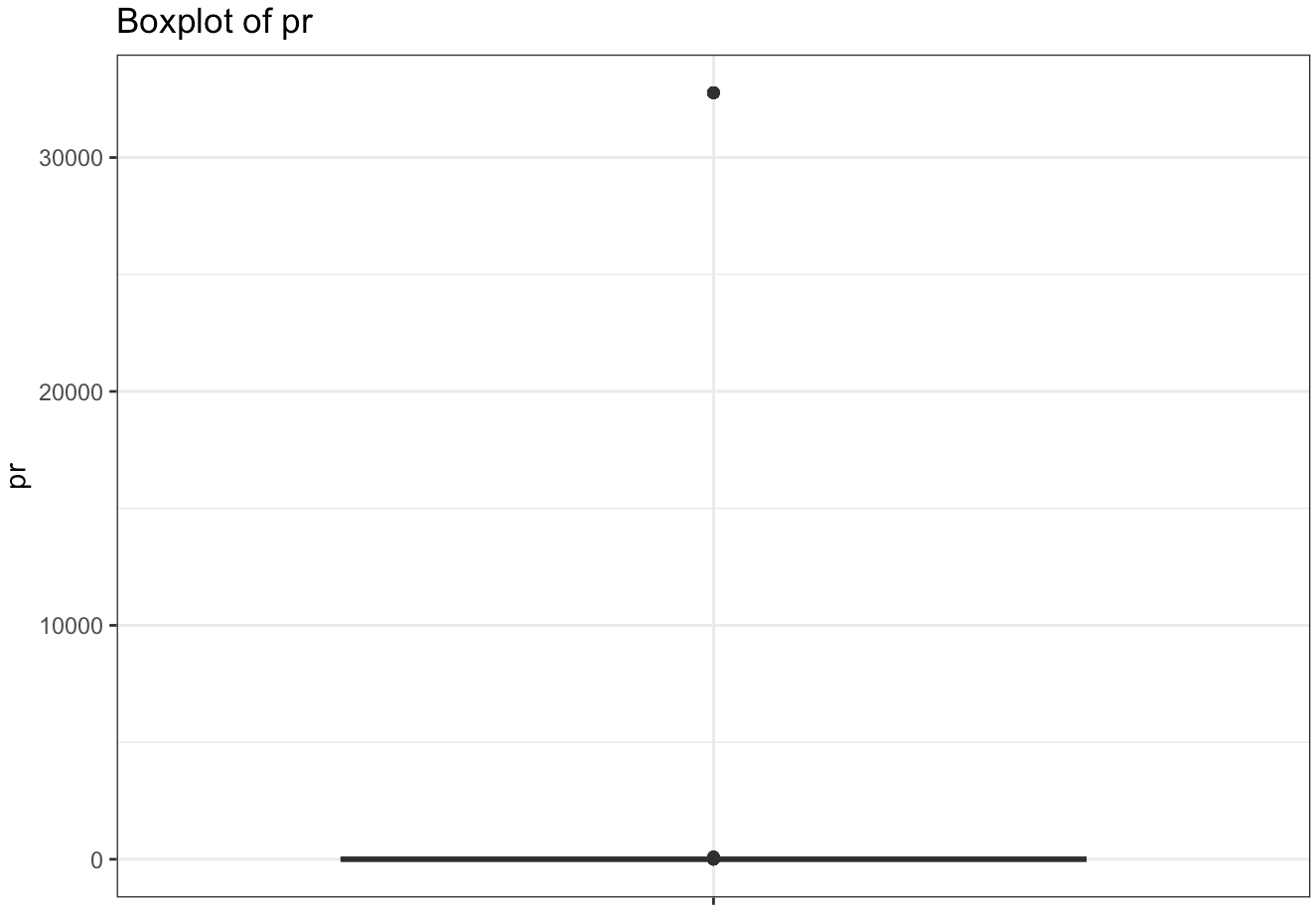
```
#anyNA(wildfires) -> false
#sum(is.na(wildfires))
```

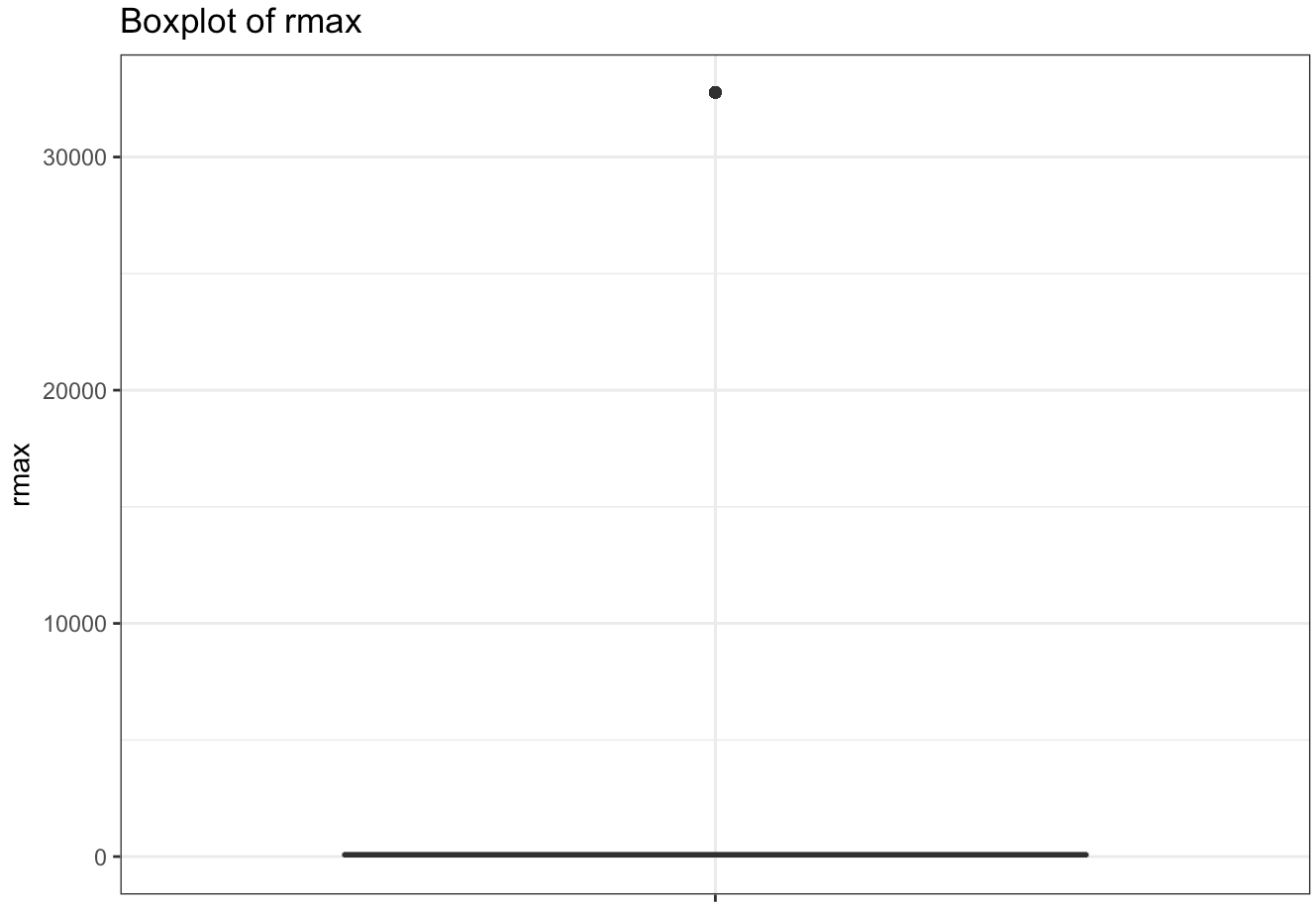
```
int_vars <- wildfires %>% select(where(is.numeric)) %>% names()

for (var in int_vars) {
  p <- ggplot(wildfires, aes(x = "", y = .data[[var]])) +
    geom_boxplot(fill = "steelblue", alpha = 0.7) +
    labs(title = paste("Boxplot of", var), x = NULL, y = var) +
    theme_bw()

  print(p)
}
```

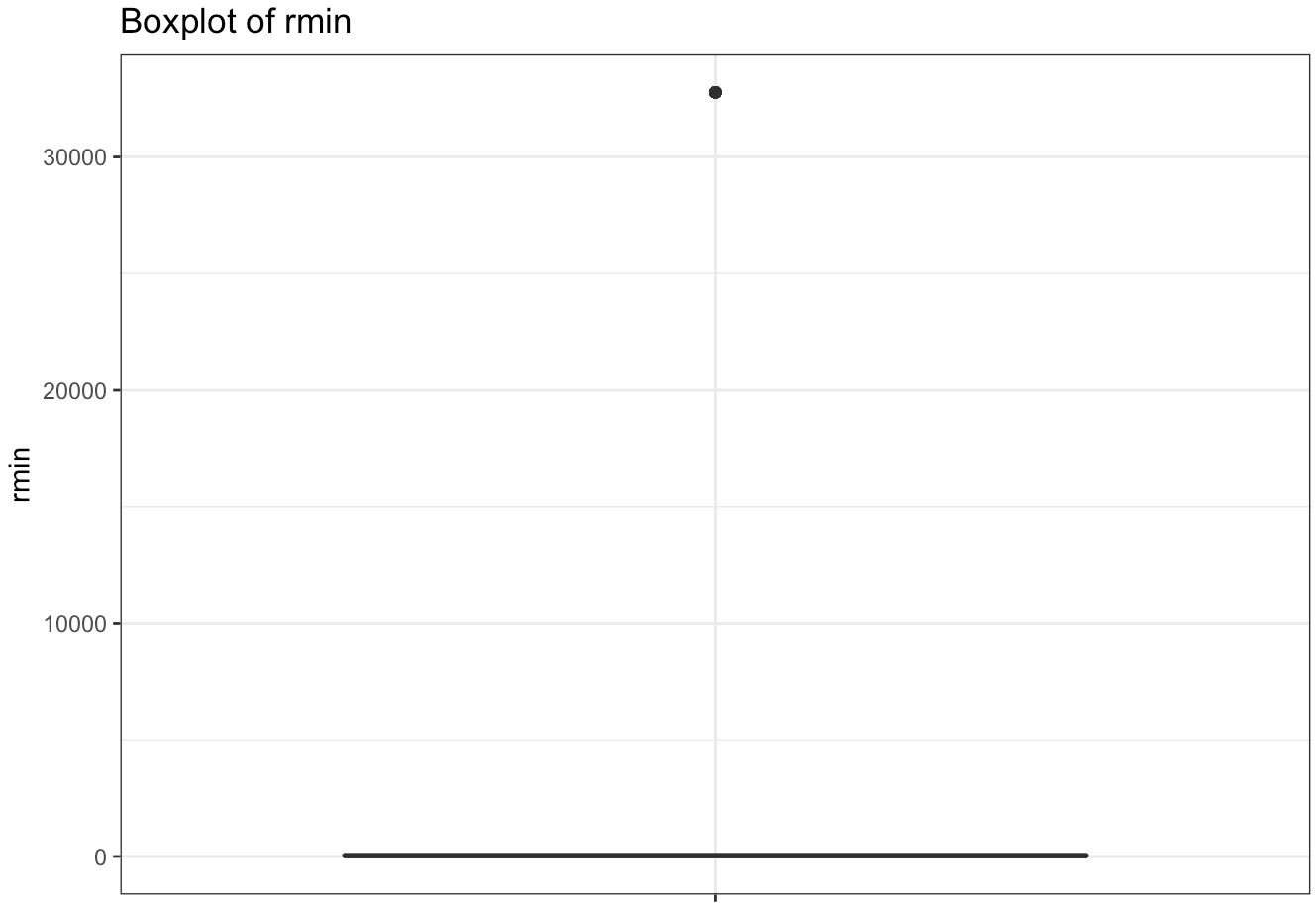
Boxplot of ...1
7500000
5000000
2500000
0
...1

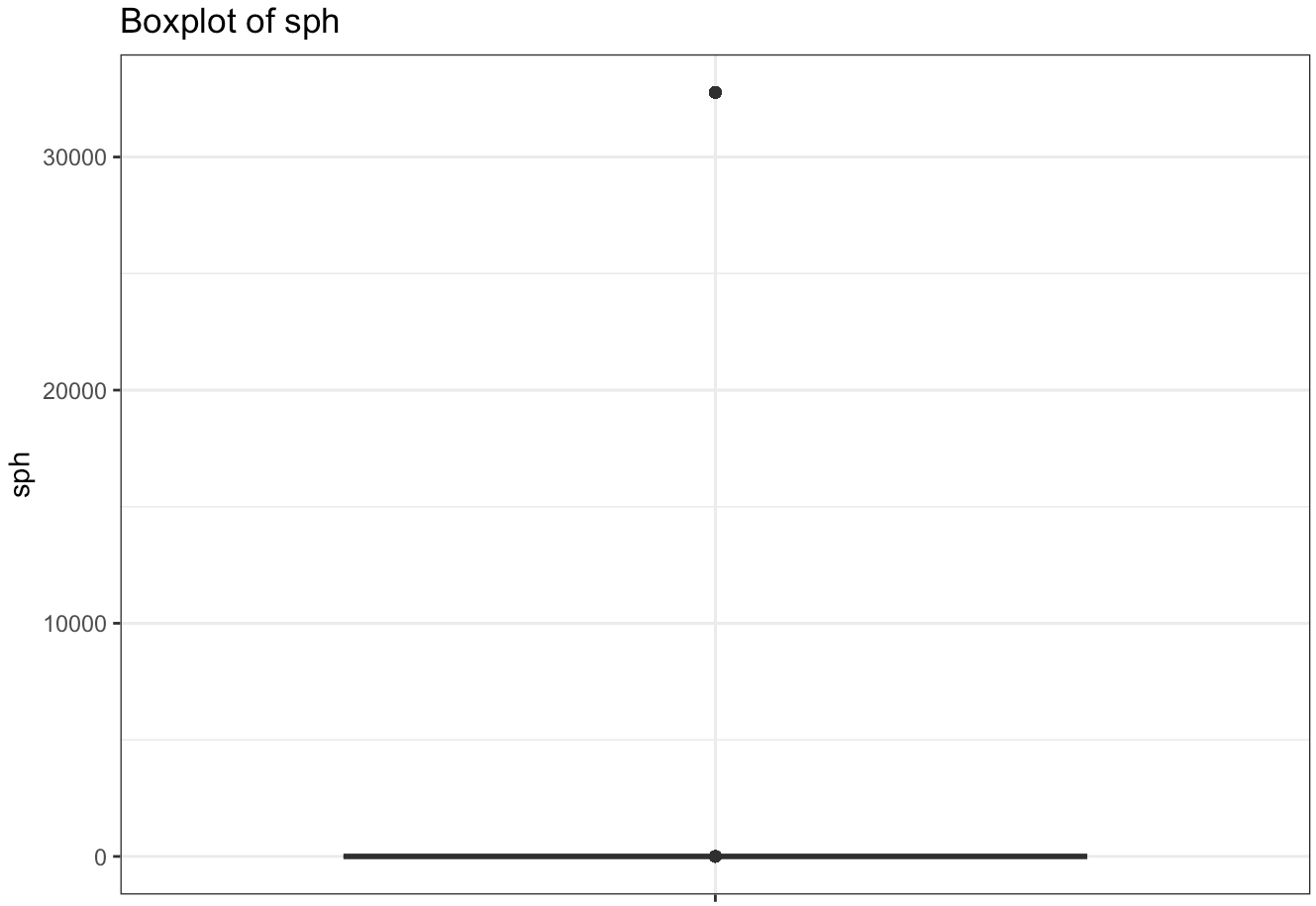

Boxplot of latitude
latitude
45
40
35
30

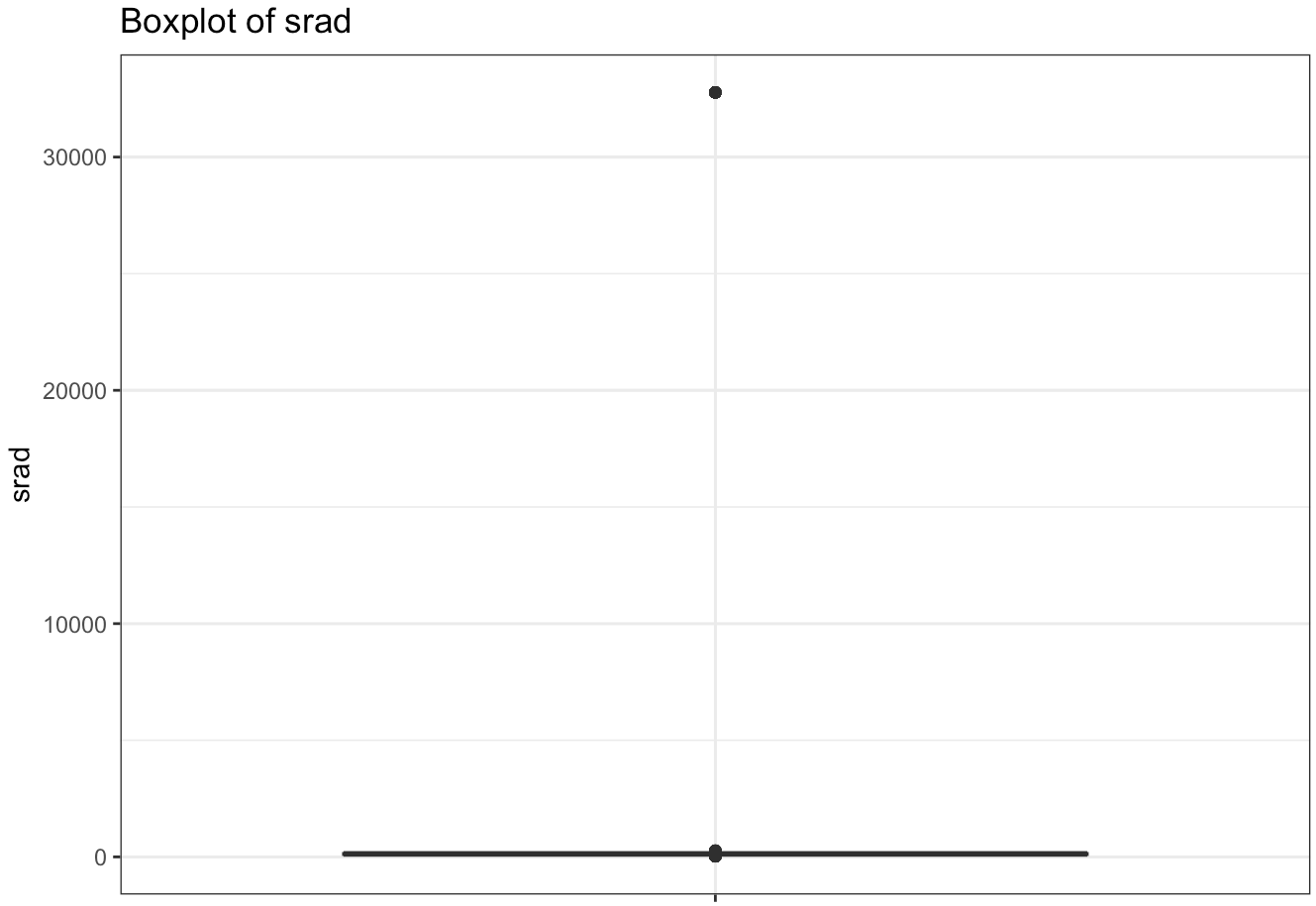

Boxplot of longitude
longitude
-70
-80
-90
-100
-110
-120

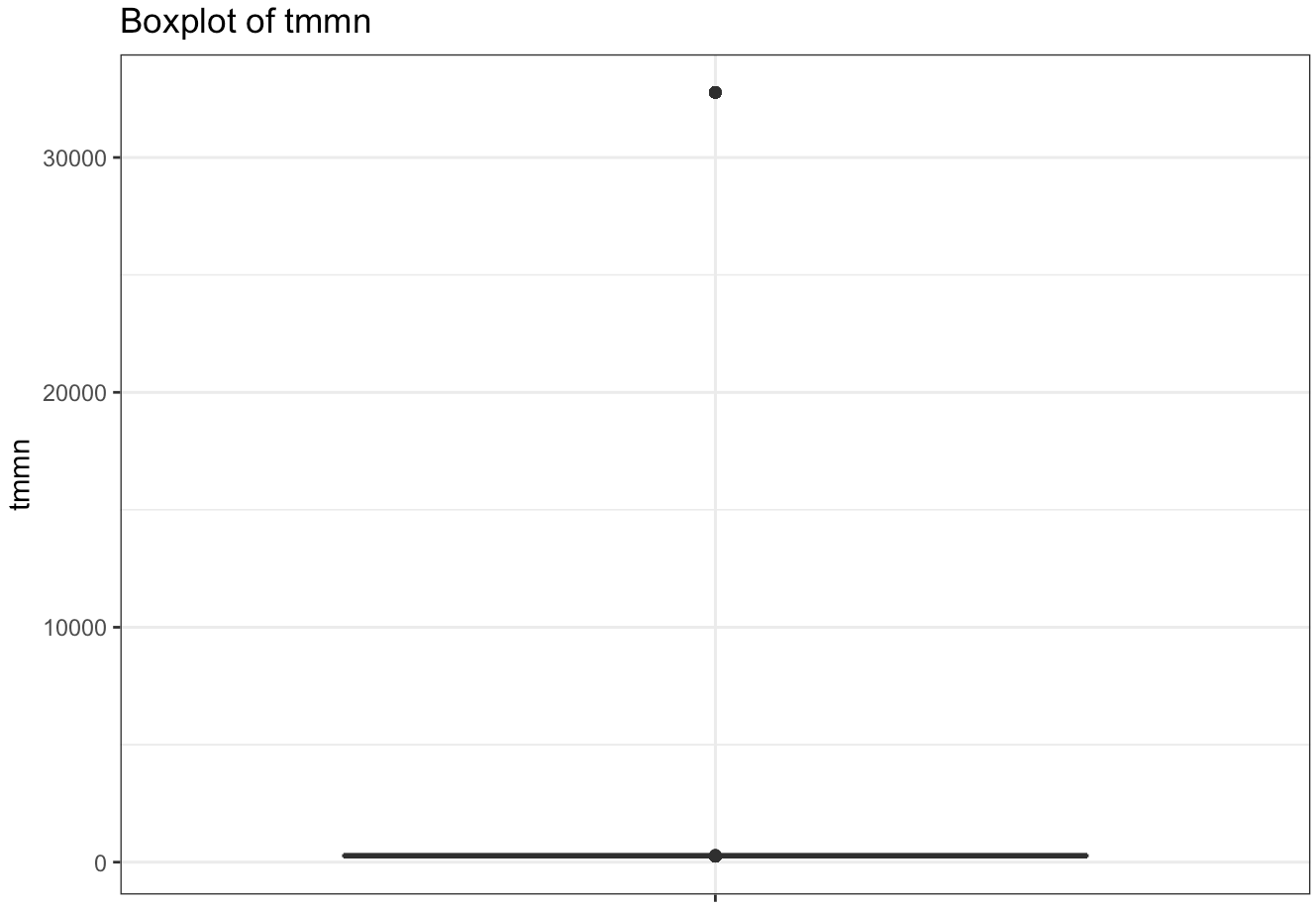

Boxplot of pr
pr
30000
20000
10000
0

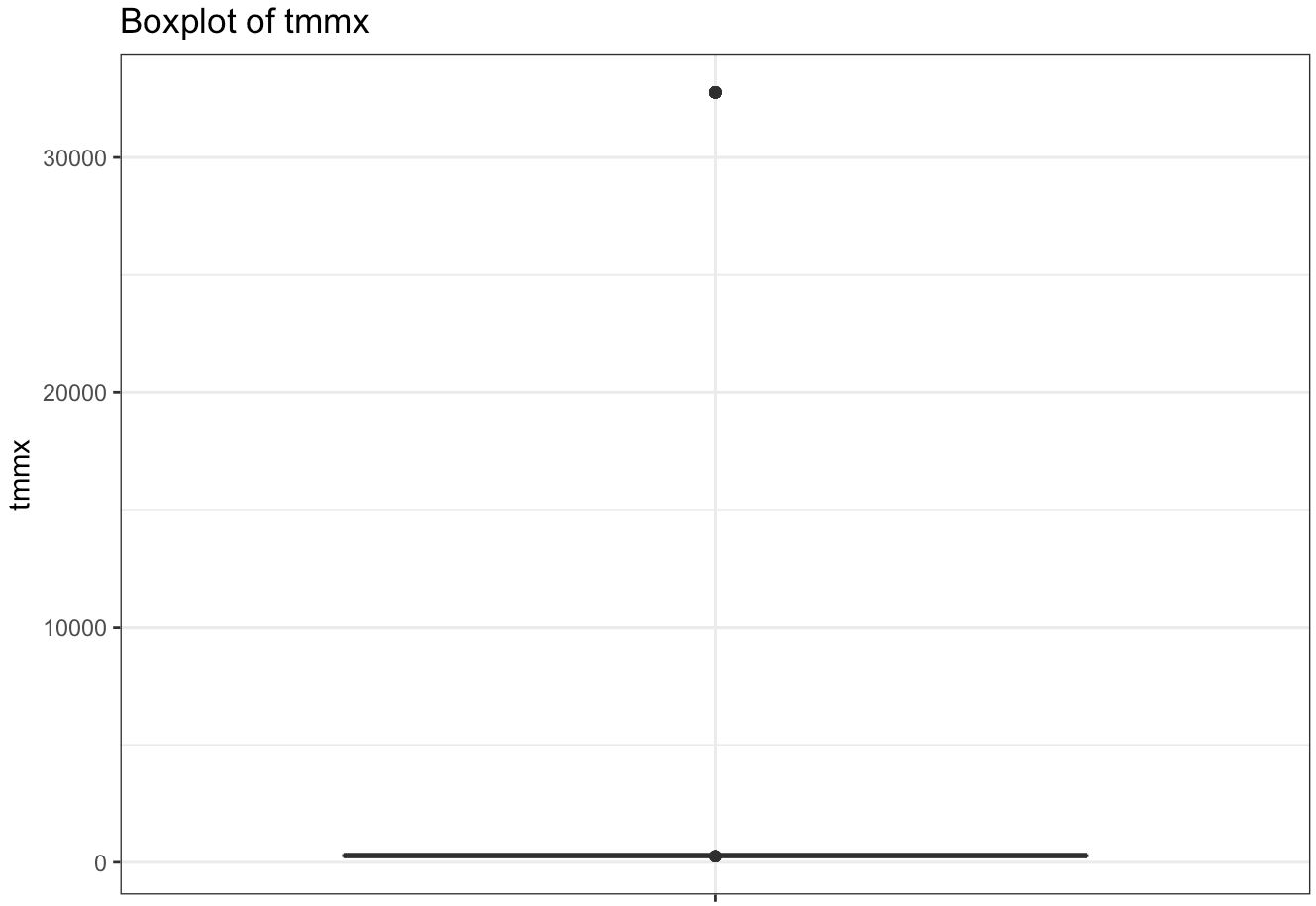

Boxplot of rmax
rmax
30000
20000
10000
0

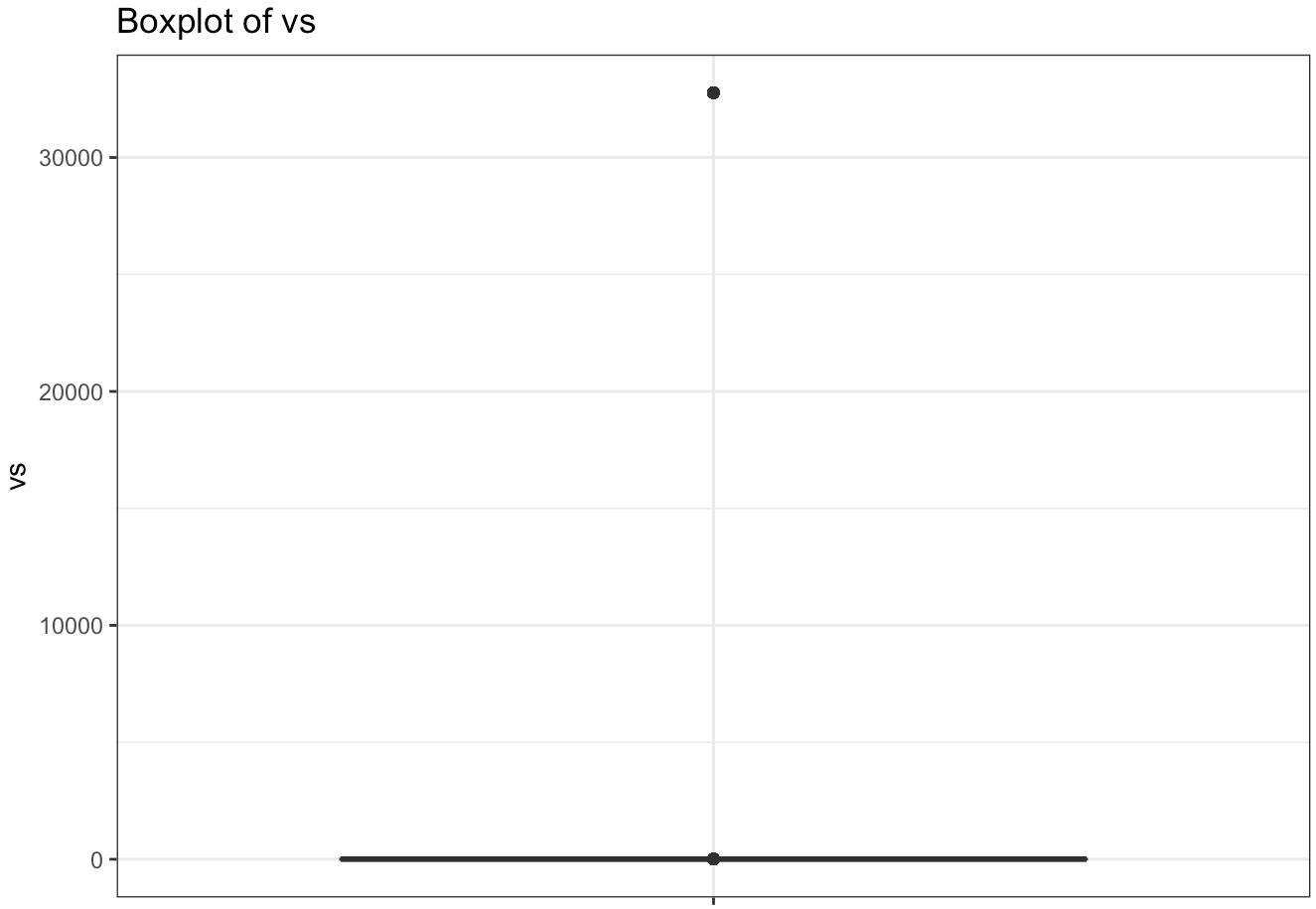

Boxplot of rmin
rmin
30000
20000
10000
0

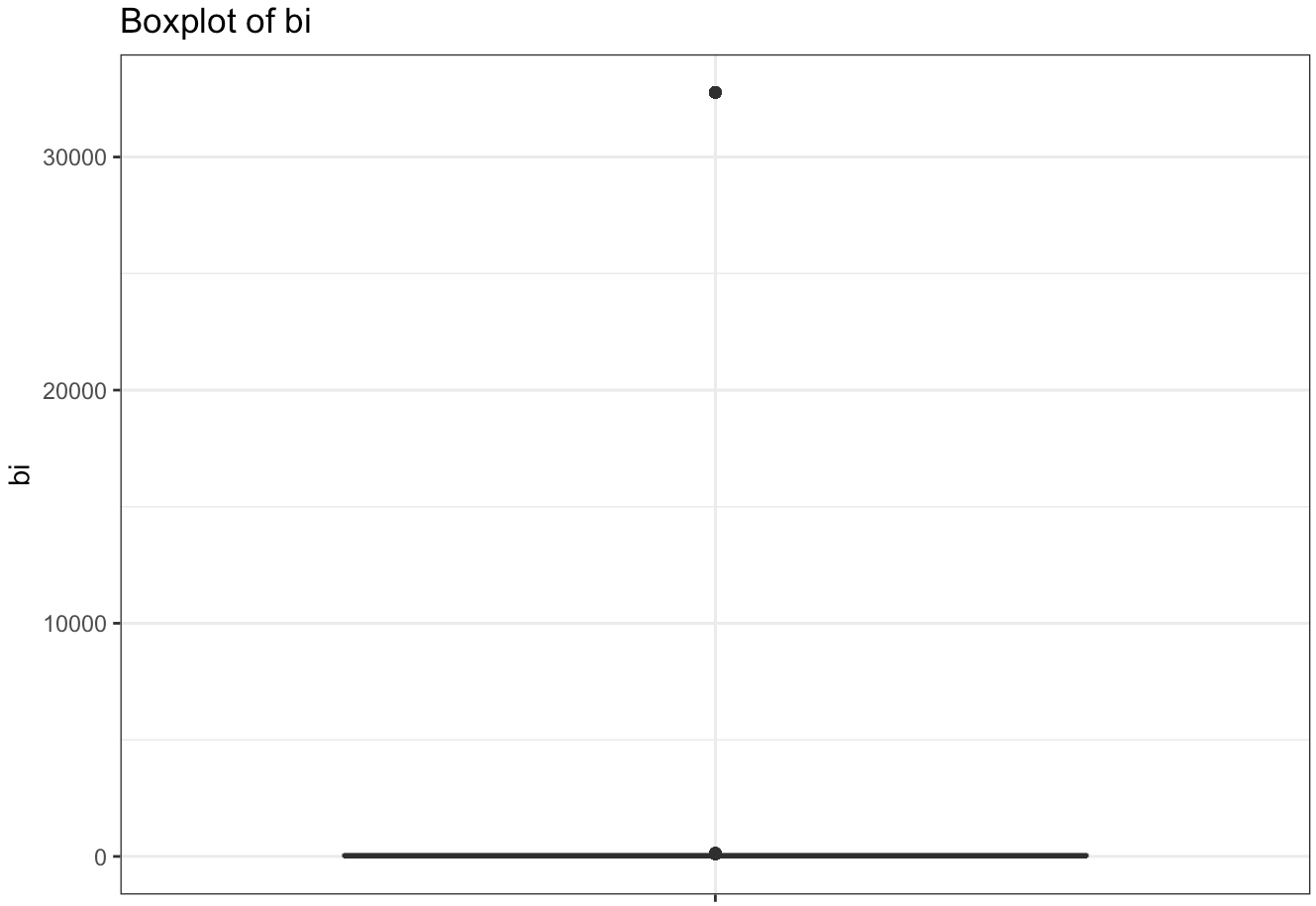

Boxplot of sph
sph
30000
20000
10000
0

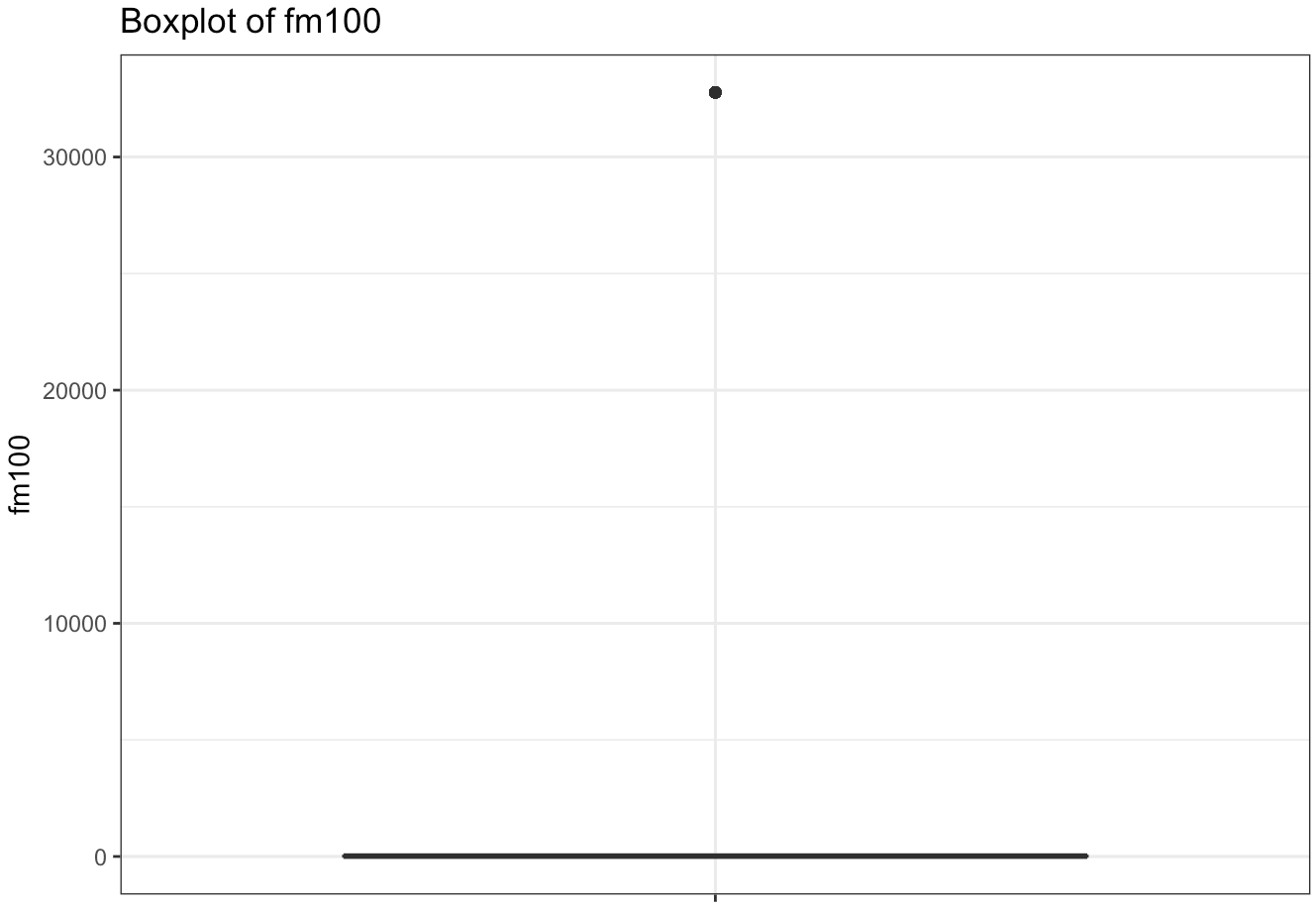

Boxplot of srad
srad
30000
20000
10000
0

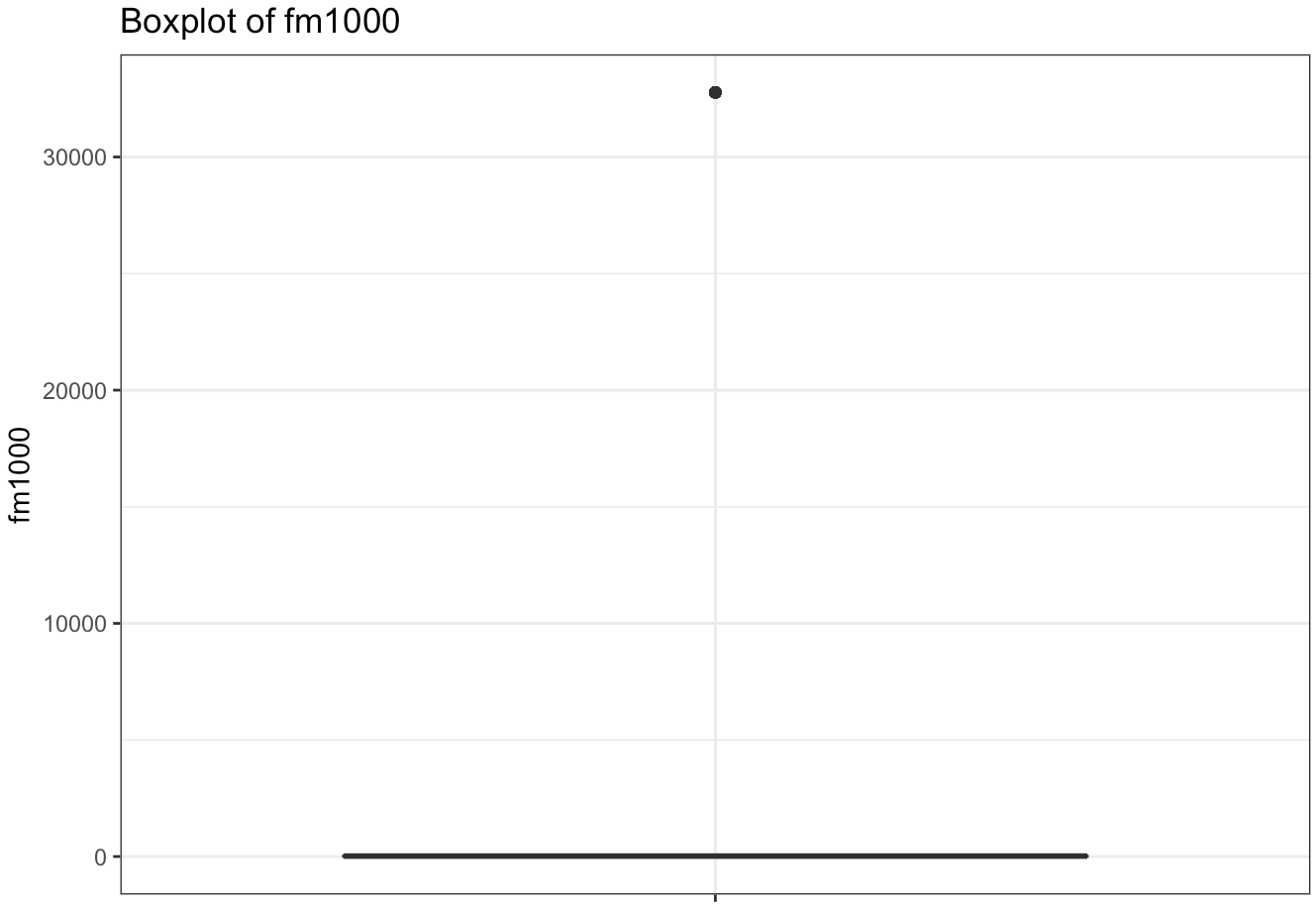

Boxplot of tmmn
tmmn
30000
20000
10000
0

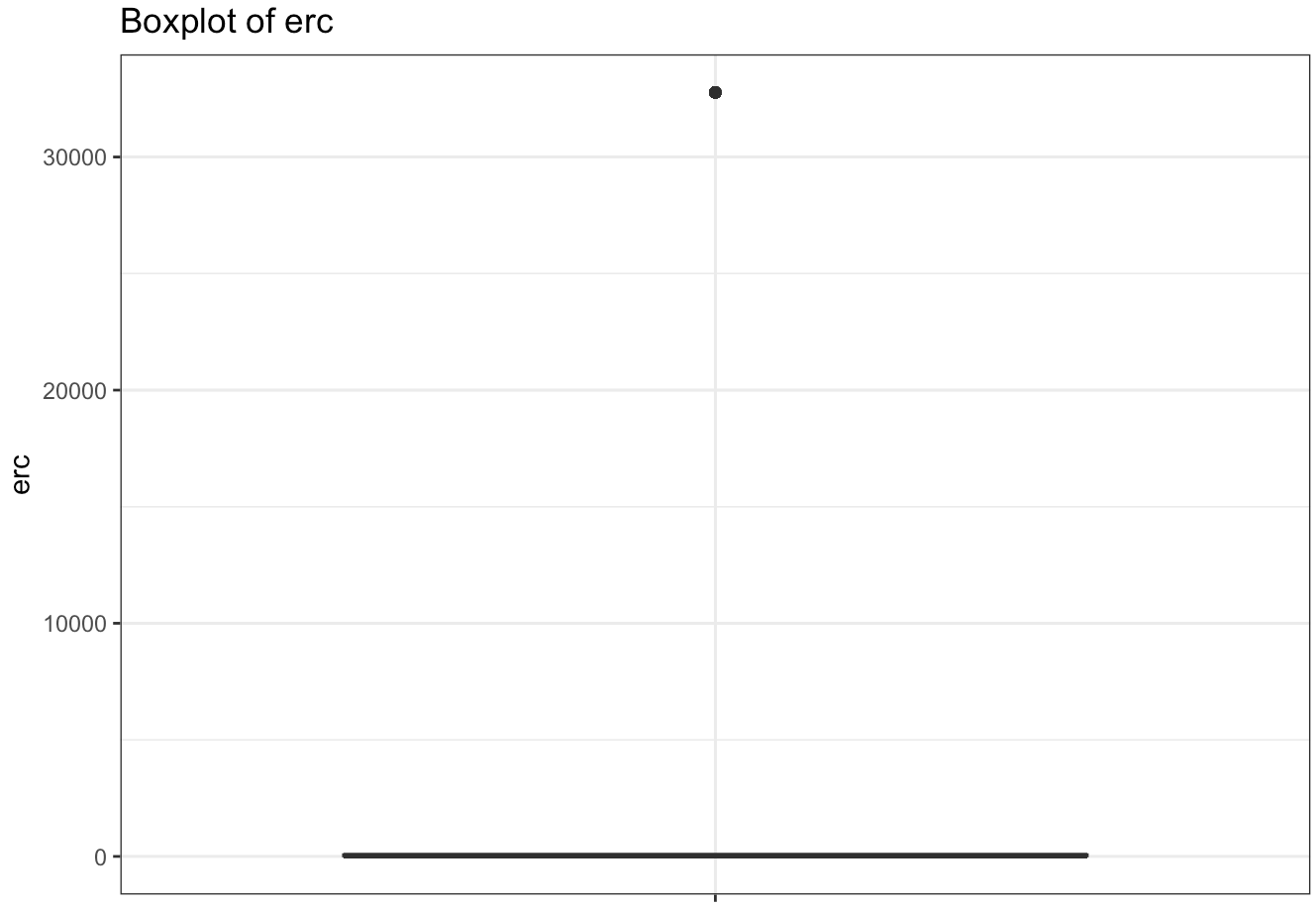

Boxplot of tmmx
tmmx
30000
20000
10000
0

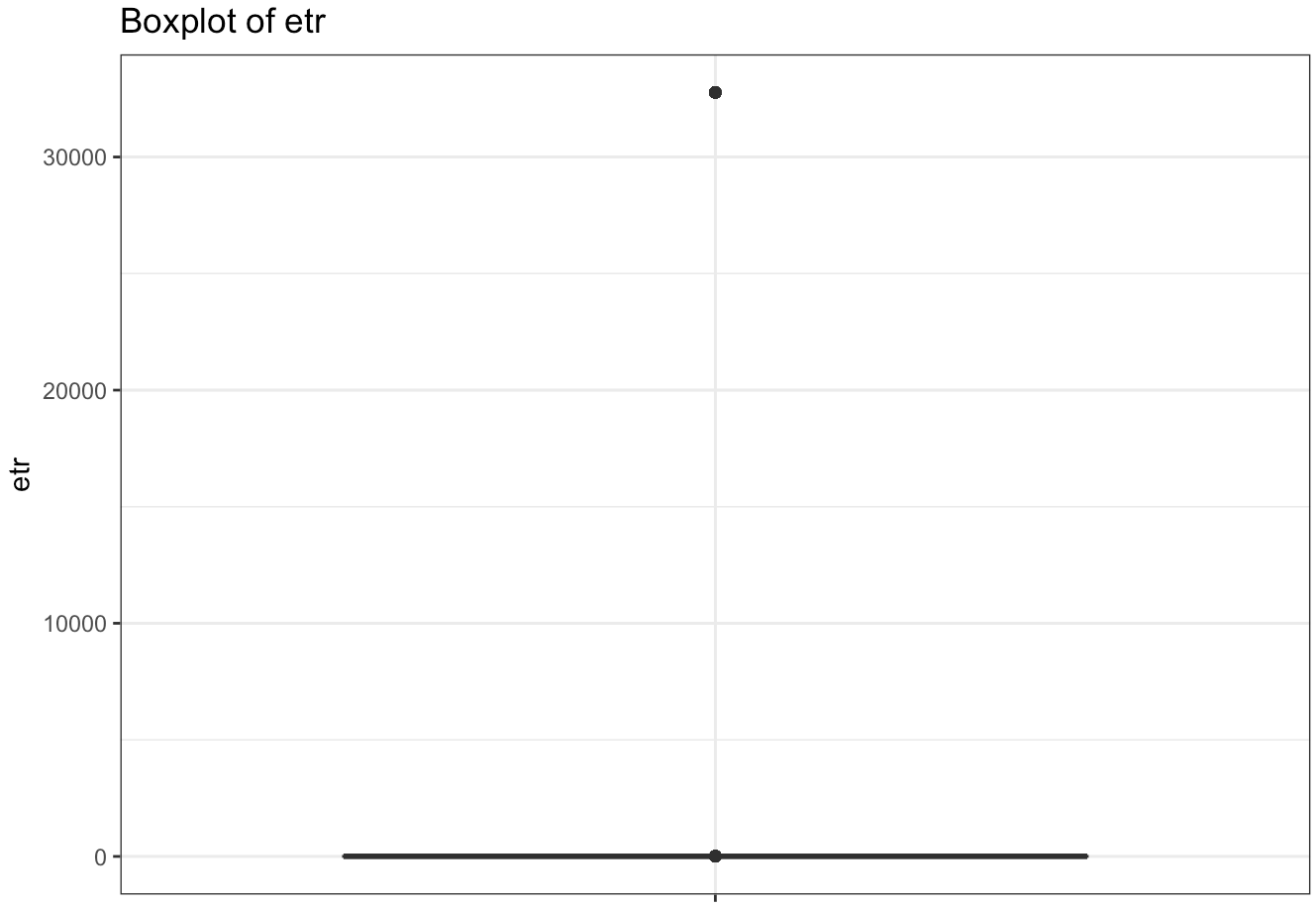

Boxplot of vs
vs
30000
20000
10000
0

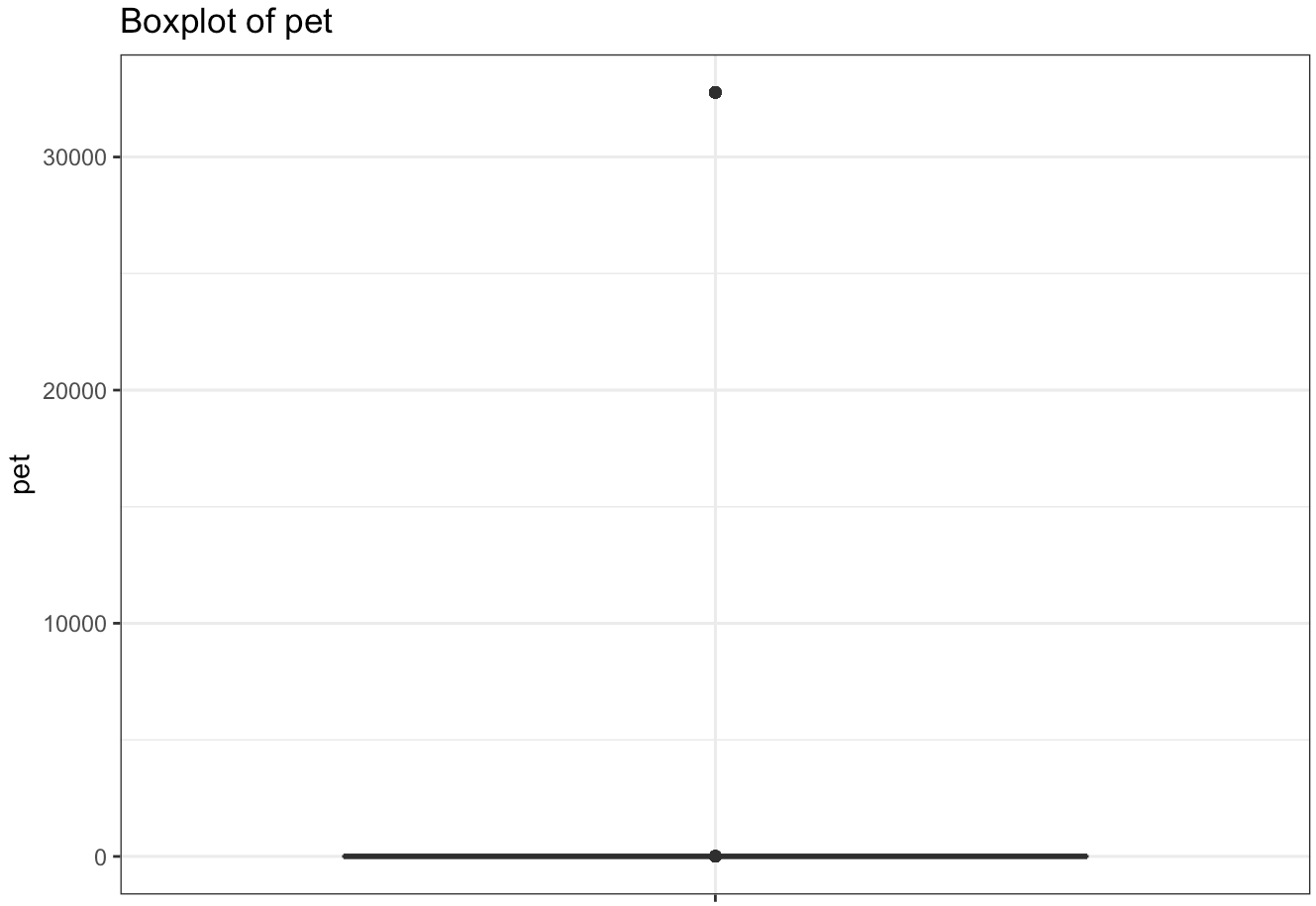

Boxplot of bi
bi
30000
20000
10000
0

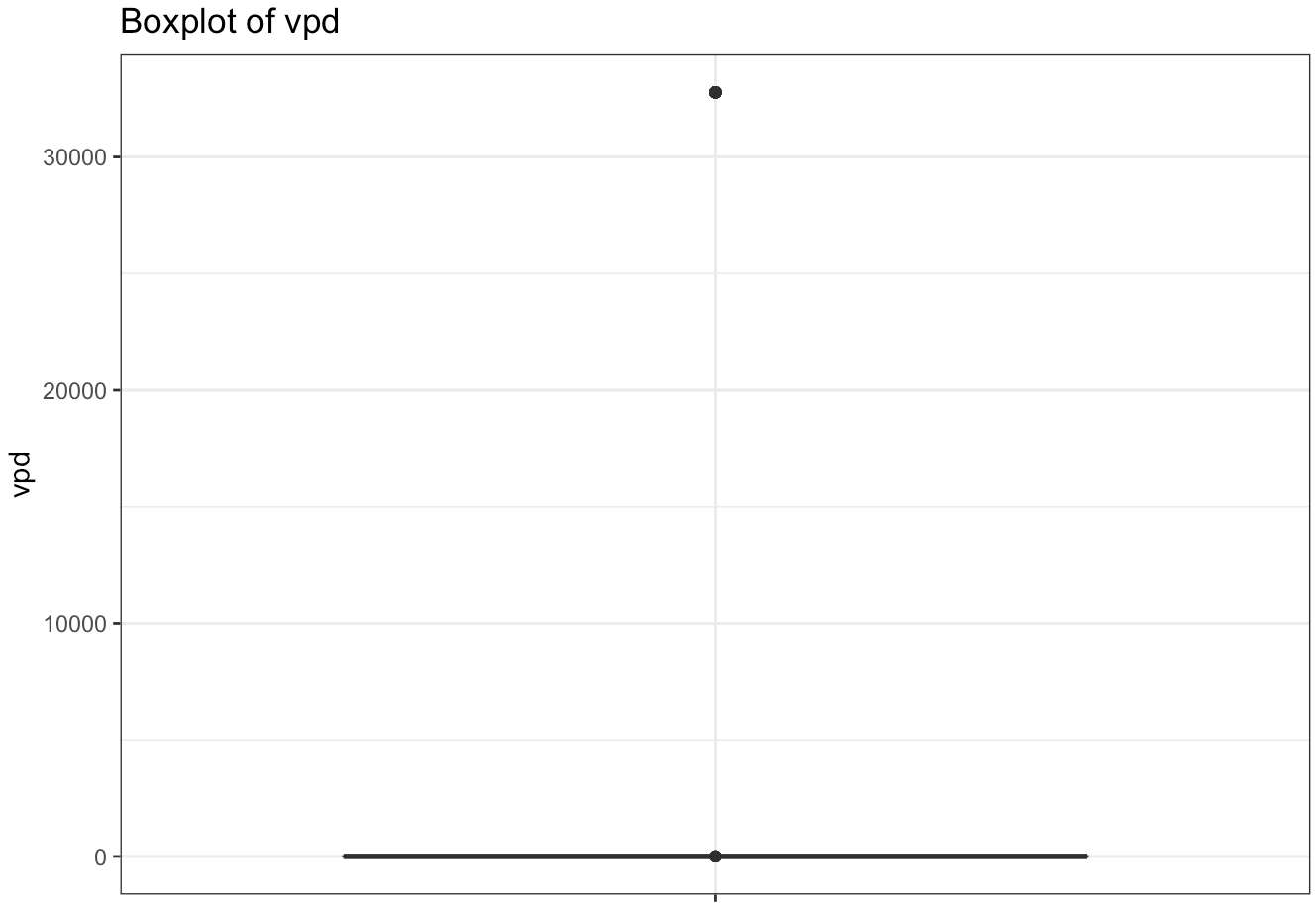

Boxplot of fm100
fm100
30000
20000
10000
0


Boxplot of fm1000
fm1000
30000
20000
10000
0


Boxplot of erc
erc
30000
20000
10000
0


Boxplot of etr
etr
30000
20000
10000
0


Boxplot of pet
pet
30000
20000
10000
0




```r
summary(wildfires %>% select(bi, vpd, tmmx, pr, fm100, rmin, vs, srad))
```

```
      bi             vpd                tmmx               pr
 Min.   :    0   Min.   :    0.00   Min.   :  250.6   Min.   :    0
 1st Qu.:   21   1st Qu.:    0.23   1st Qu.:  278.0   1st Qu.:    0
 Median :   30   Median :    0.57   Median :  287.8   Median :    0
 Mean   : 1227   Mean   : 1192.92   Mean   : 1466.6   Mean   : 1194
 3rd Qu.:   56   3rd Qu.:    1.03   3rd Qu.:  292.2   3rd Qu.:    0
 Max.   :32767   Max.   :32767.00   Max.   :32767.0   Max.   :32767
    fm100              rmin               vs               srad
 Min.   :    2.7   Min.   :    1.0   Min.   :    0.5   Min.   :   21.6
```

```
 1st Qu.:    8.5   1st Qu.:   18.0   1st Qu.:    2.6   1st Qu.:  103.0
 Median :   14.9   Median :   35.4   Median :    3.8   Median :  129.5
 Mean   : 1205.3   Mean   : 1225.5   Mean   : 1196.2   Mean   : 1311.3
 3rd Qu.:   18.4   3rd Qu.:   52.1   3rd Qu.:    5.3   3rd Qu.:  150.9
 Max.   :32767.0   Max.   :32767.0   Max.   :32767.0   Max.   :32767.0
```

```r
wildfires_clean <- wildfires %>%
  filter(!if_any(everything(), ~ . == 32767))
```

```r
nrow(wildfires)        # original
```

```
[1] 19210
```

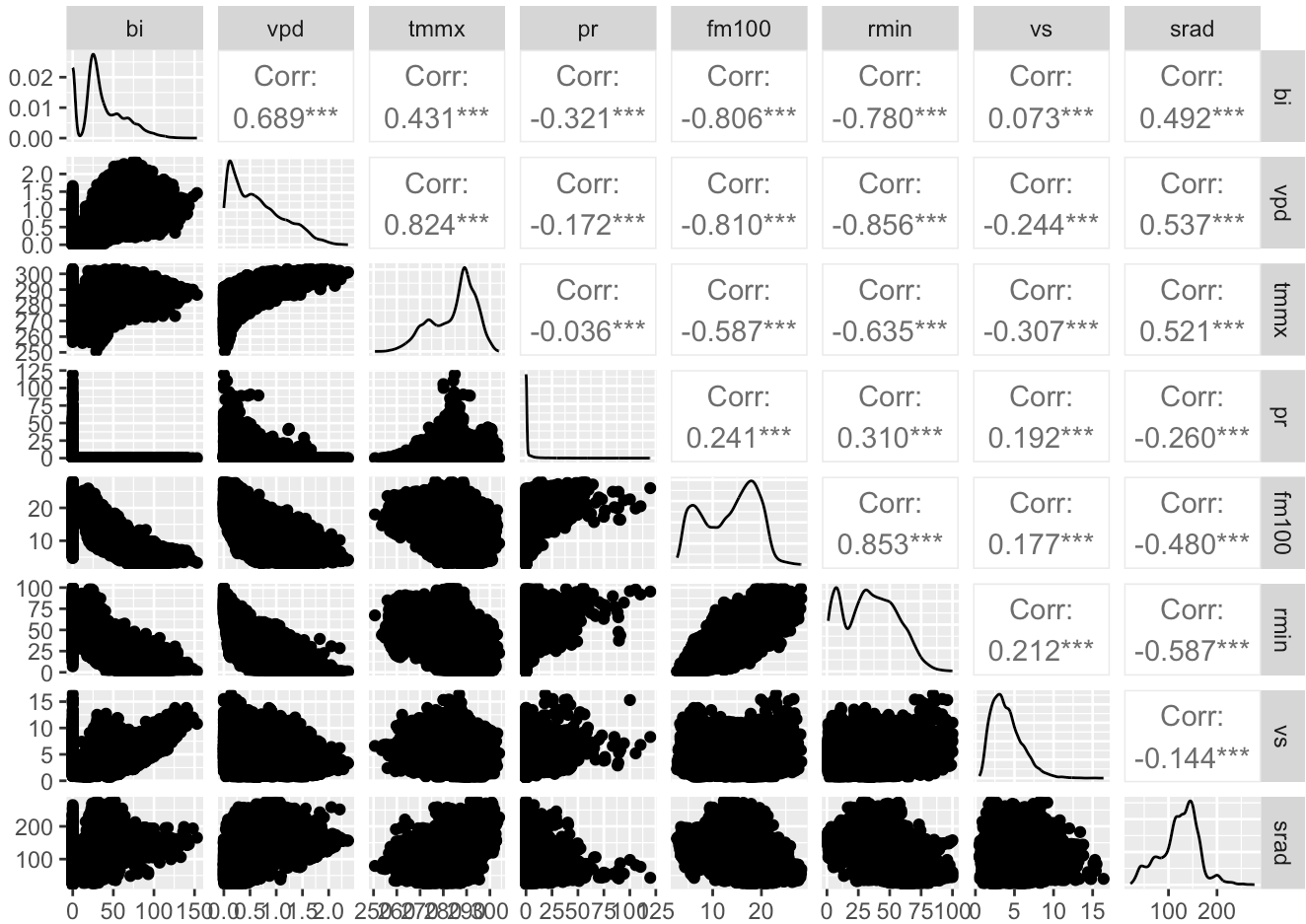
```r
nrow(wildfires_clean)  # after removing
```

```
[1] 18511
```

```r
nrow(wildfires) - nrow(wildfires_clean)  # rows dropped
```

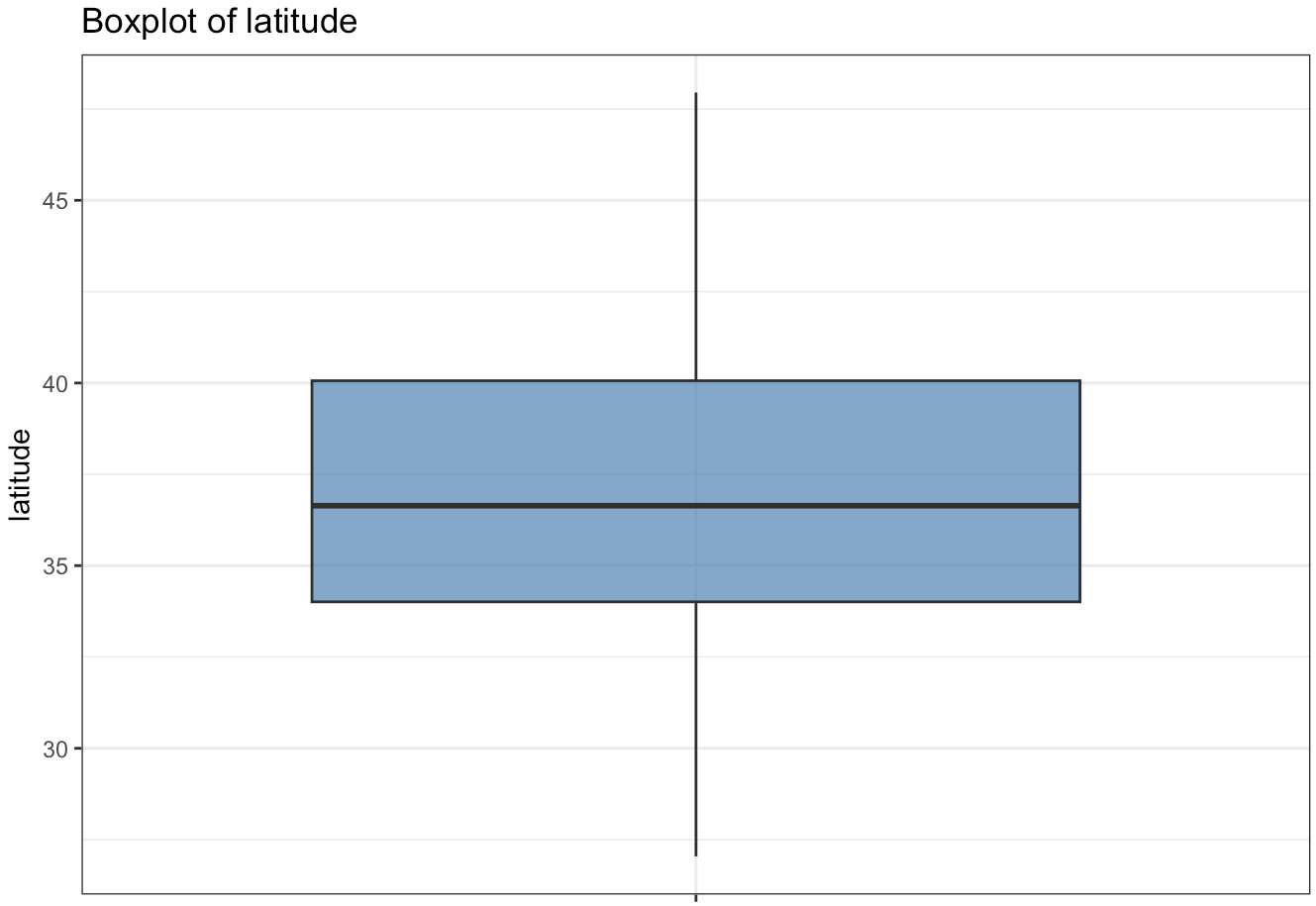
```
[1] 699
```

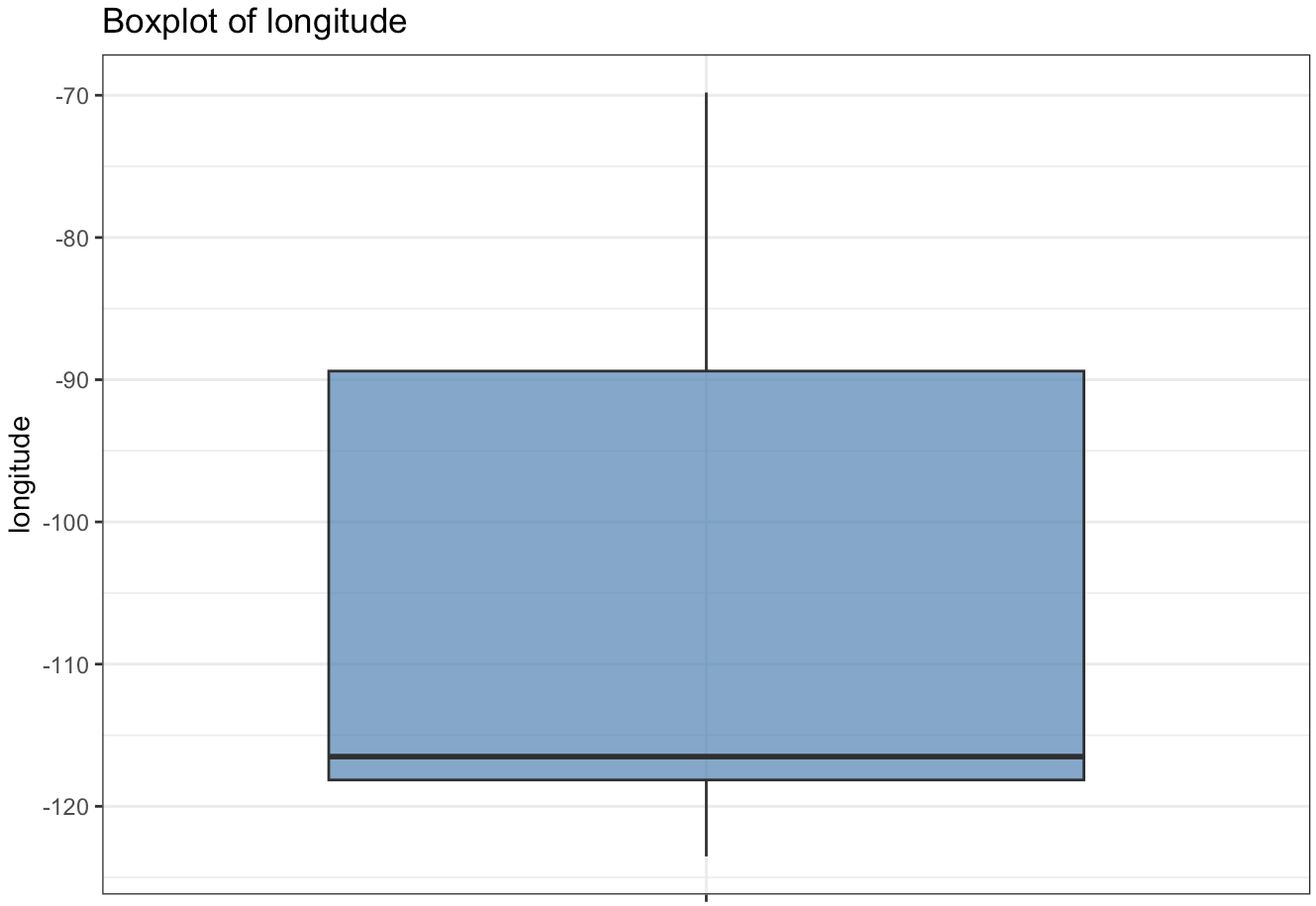
```r
wildfires_clean %>%
  select(bi, vpd, tmmx, pr, fm100, rmin, vs, srad) %>%
  ggpairs()
```

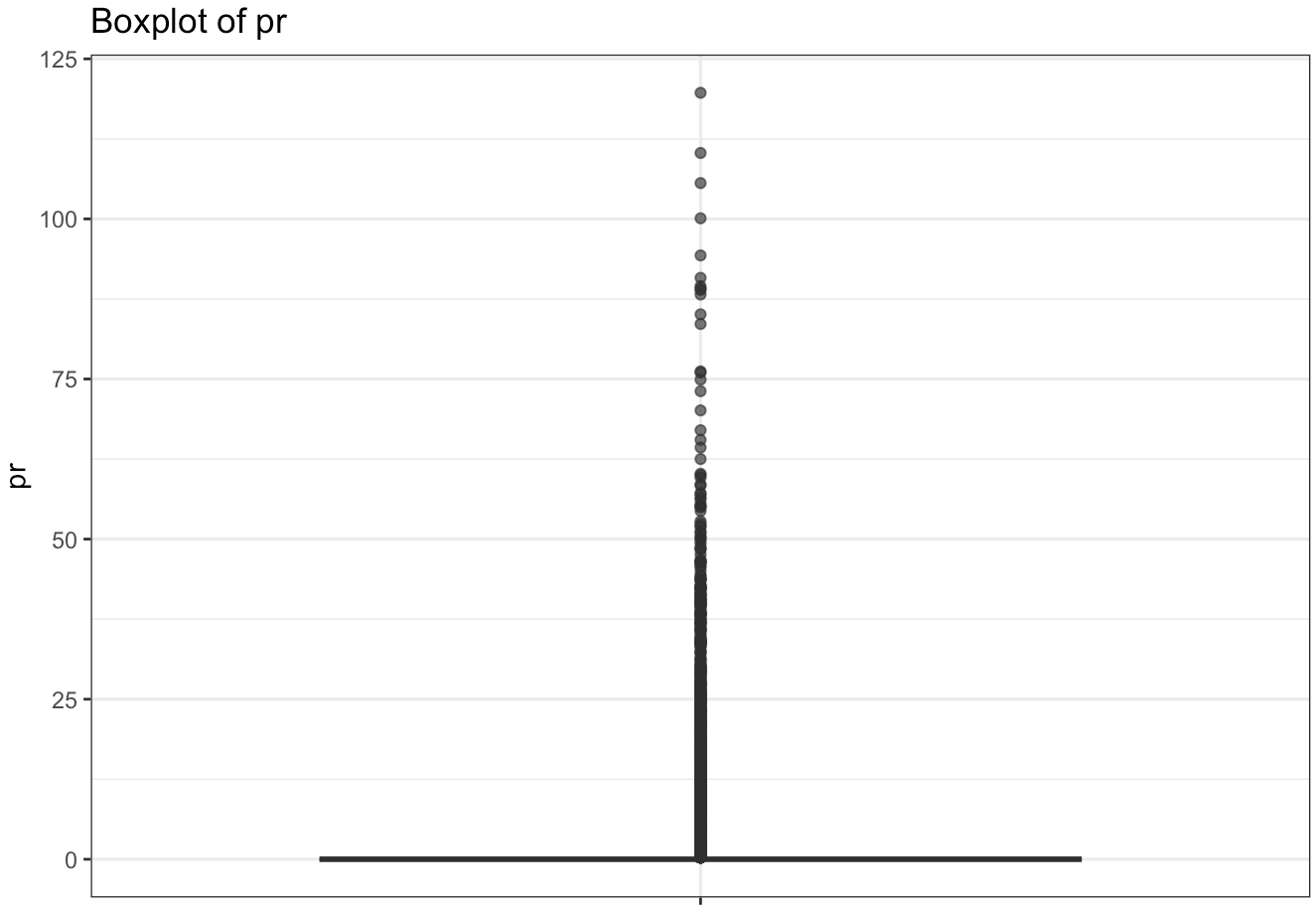


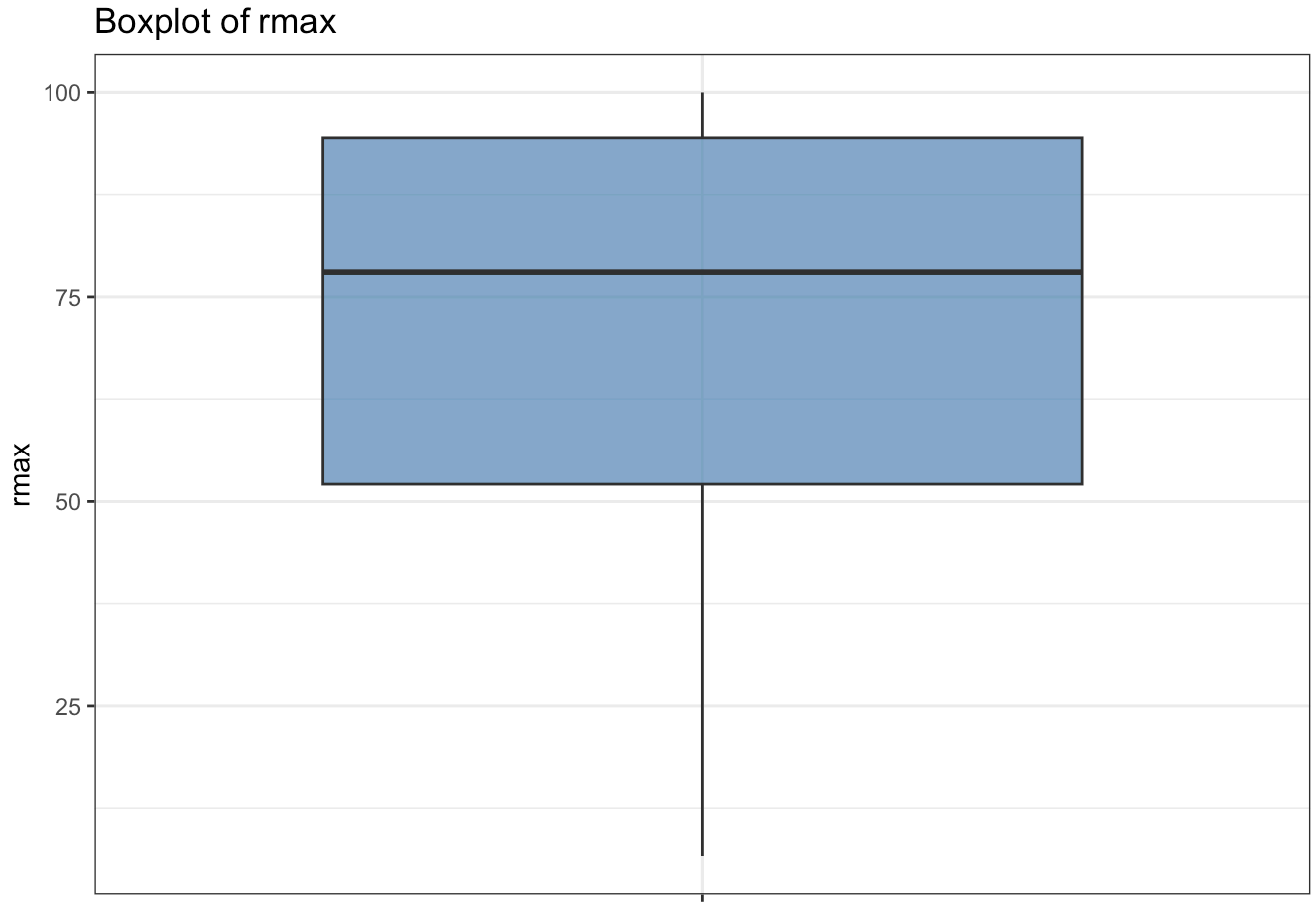

```r
int_vars <- wildfires_clean %>% select(where(is.numeric)) %>% names()

for (var in int_vars) {
  p <- ggplot(wildfires_clean, aes(x = "", y = .data[[var]])) +
    geom_boxplot(fill = "steelblue", alpha = 0.7) +
    labs(title = paste("Boxplot of", var), x = NULL, y = var) +
    theme_bw()

  print(p)
}
```

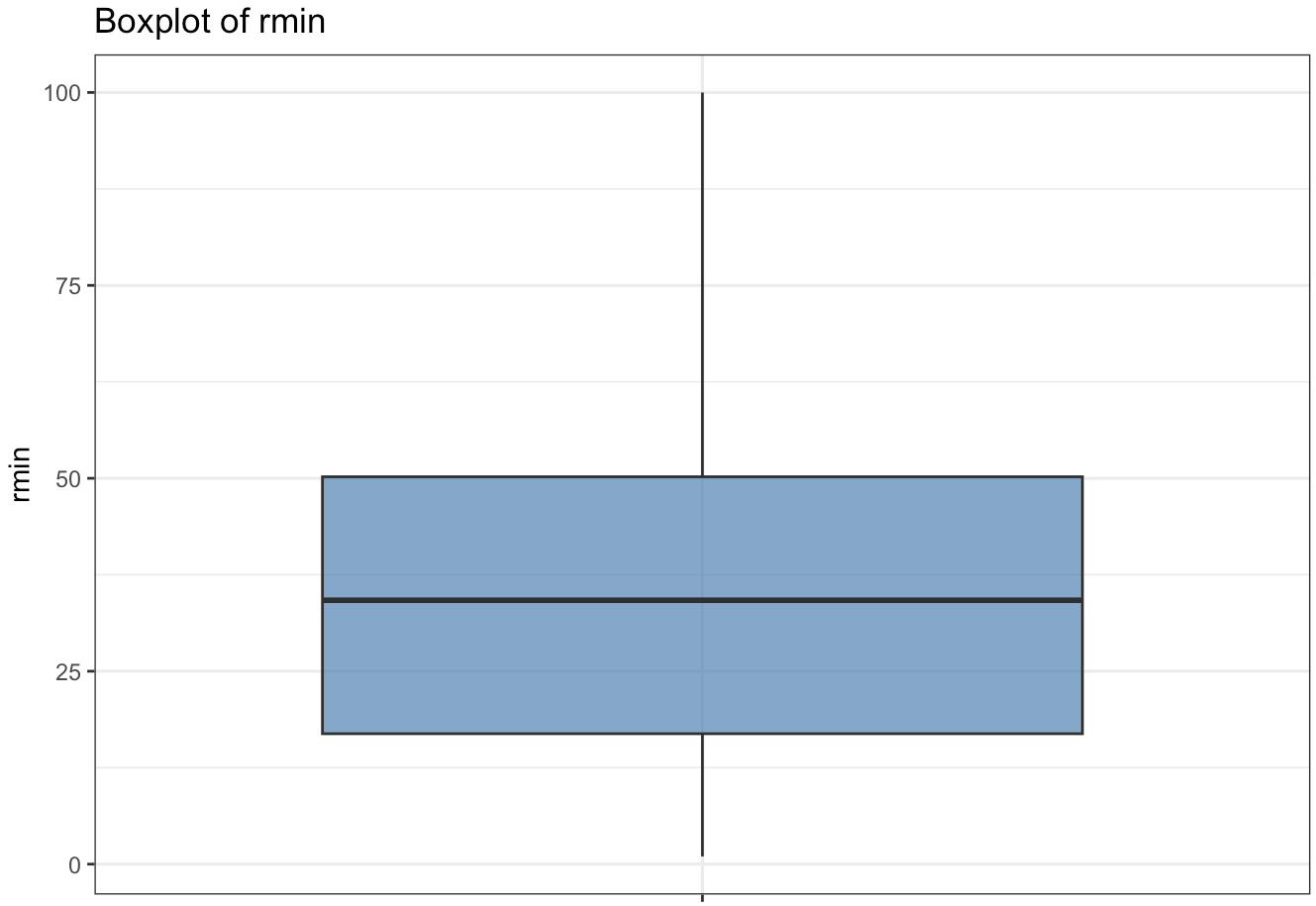
Boxplot of ...1
7500000
5000000
2500000
0
...1

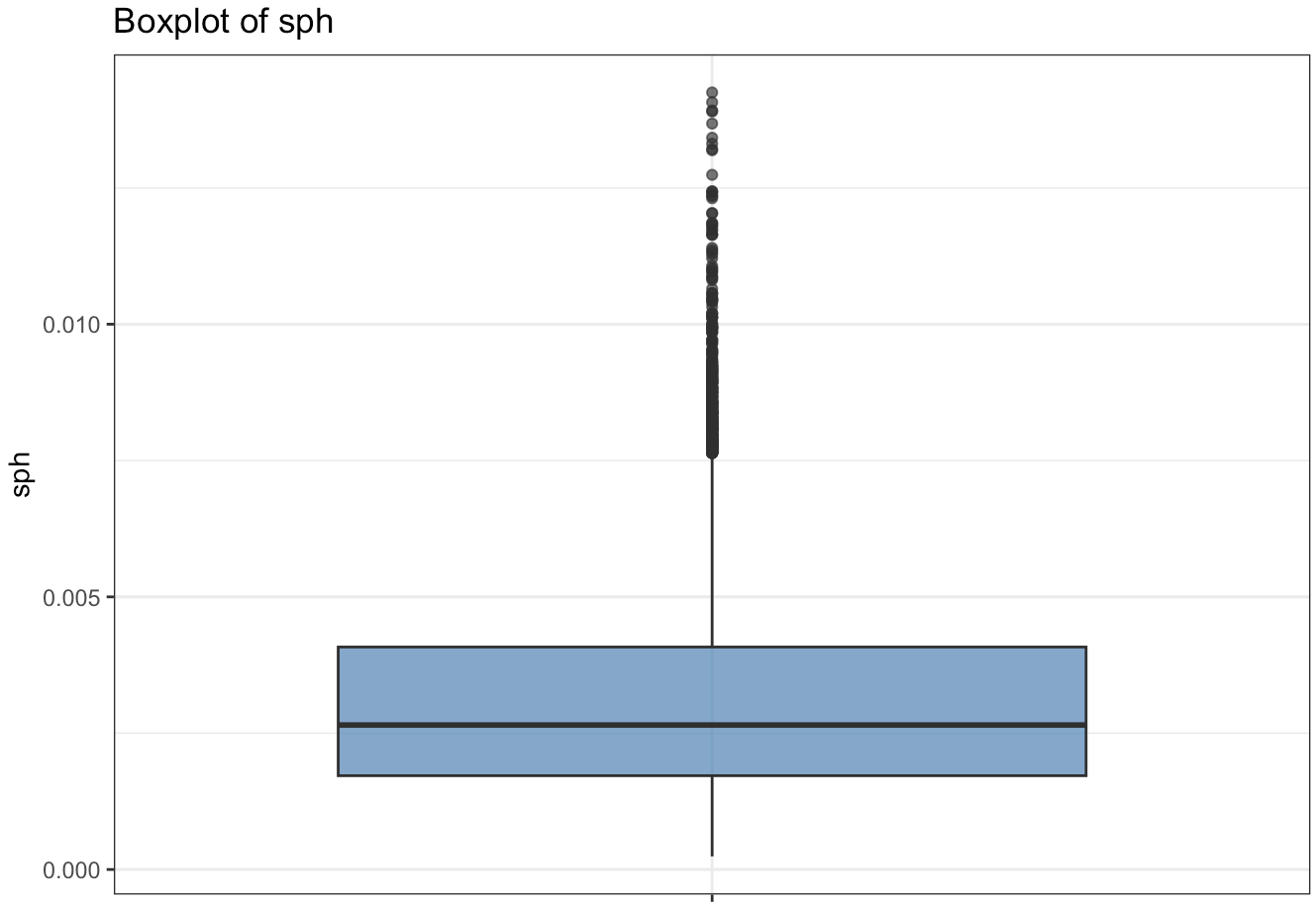

Boxplot of latitude
latitude
45
40
35
30

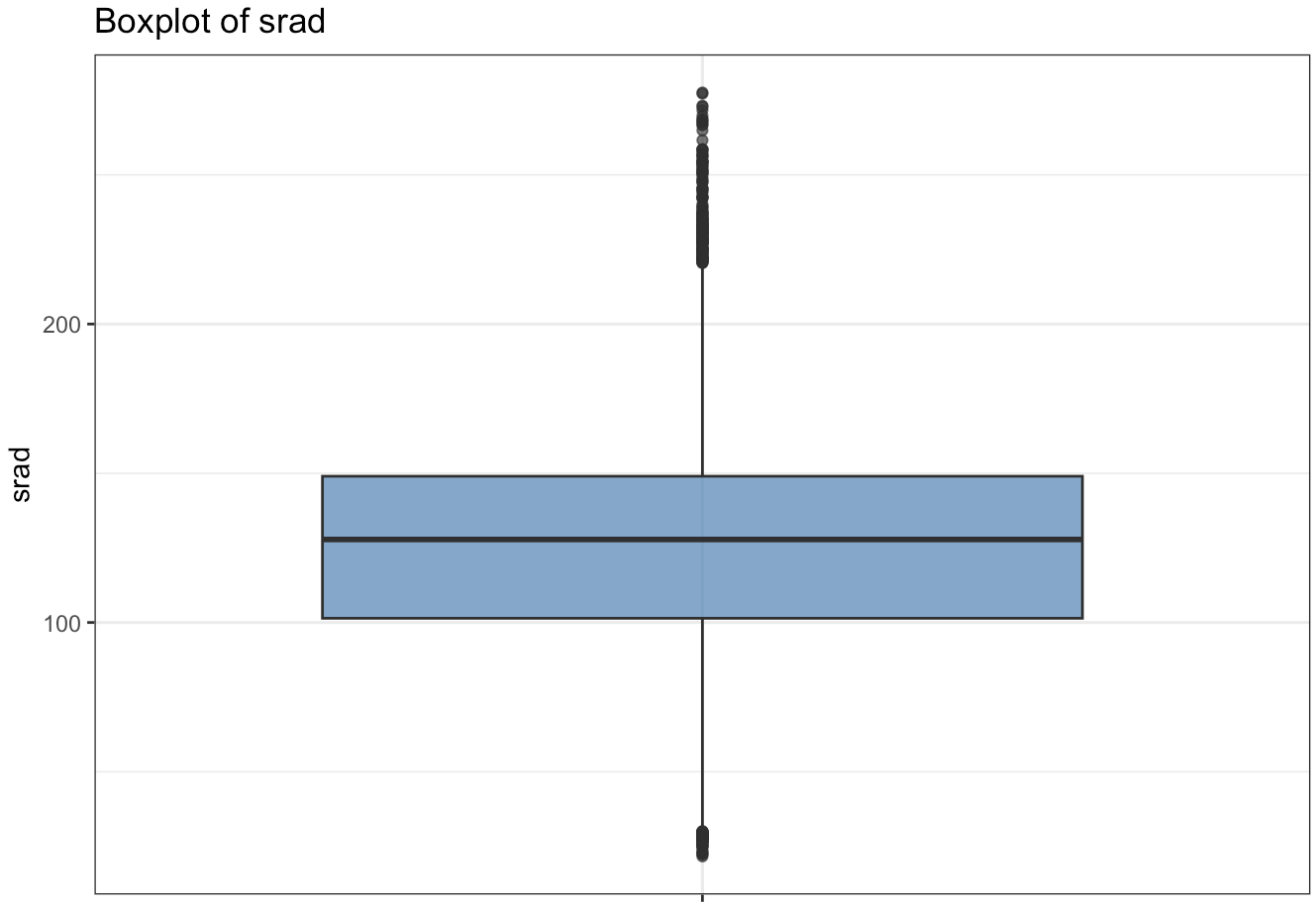

Boxplot of longitude
longitude
-70
-80
-90
-100
-110
-120

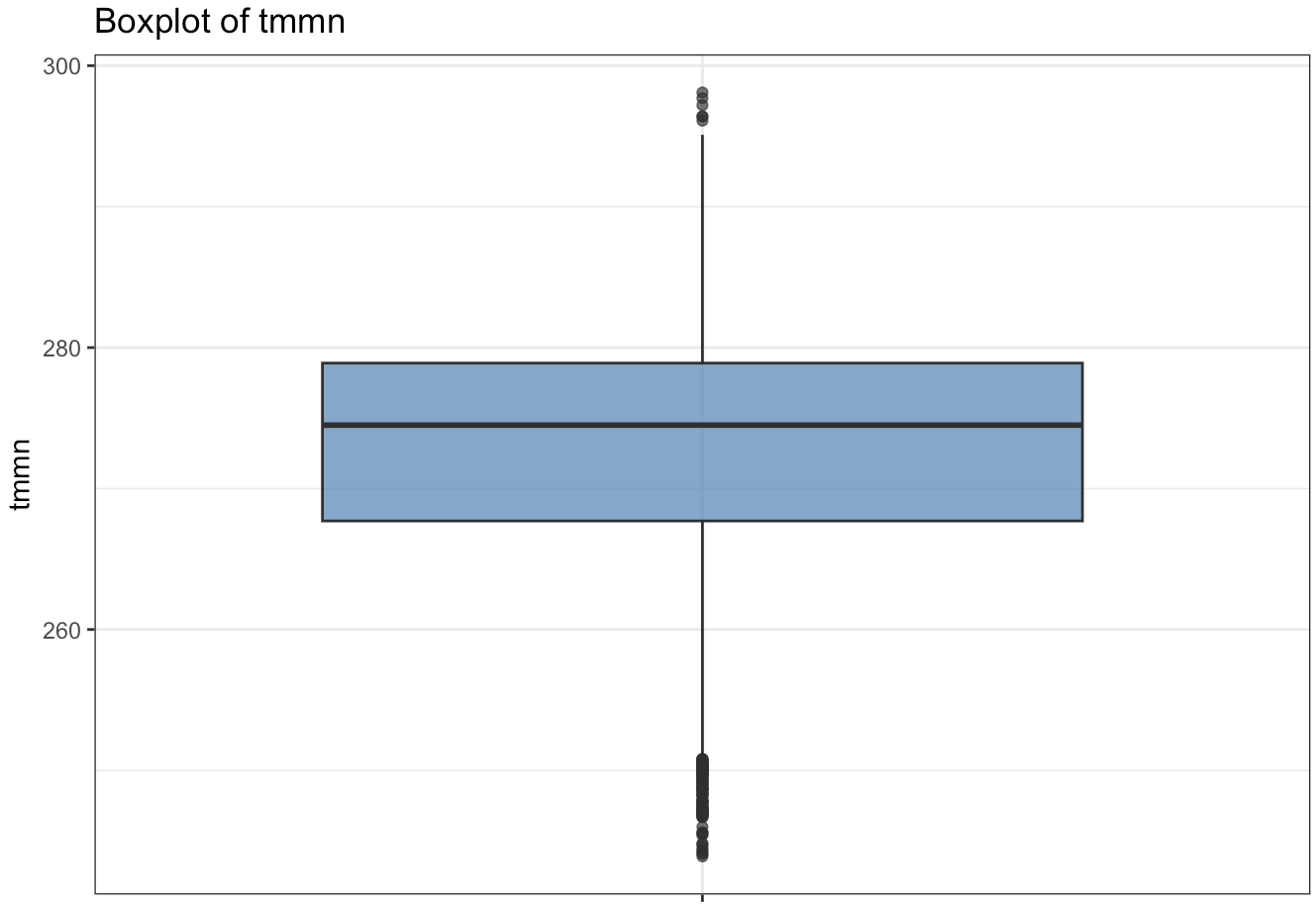

Boxplot of pr
125
100
75
50
25
0
pr

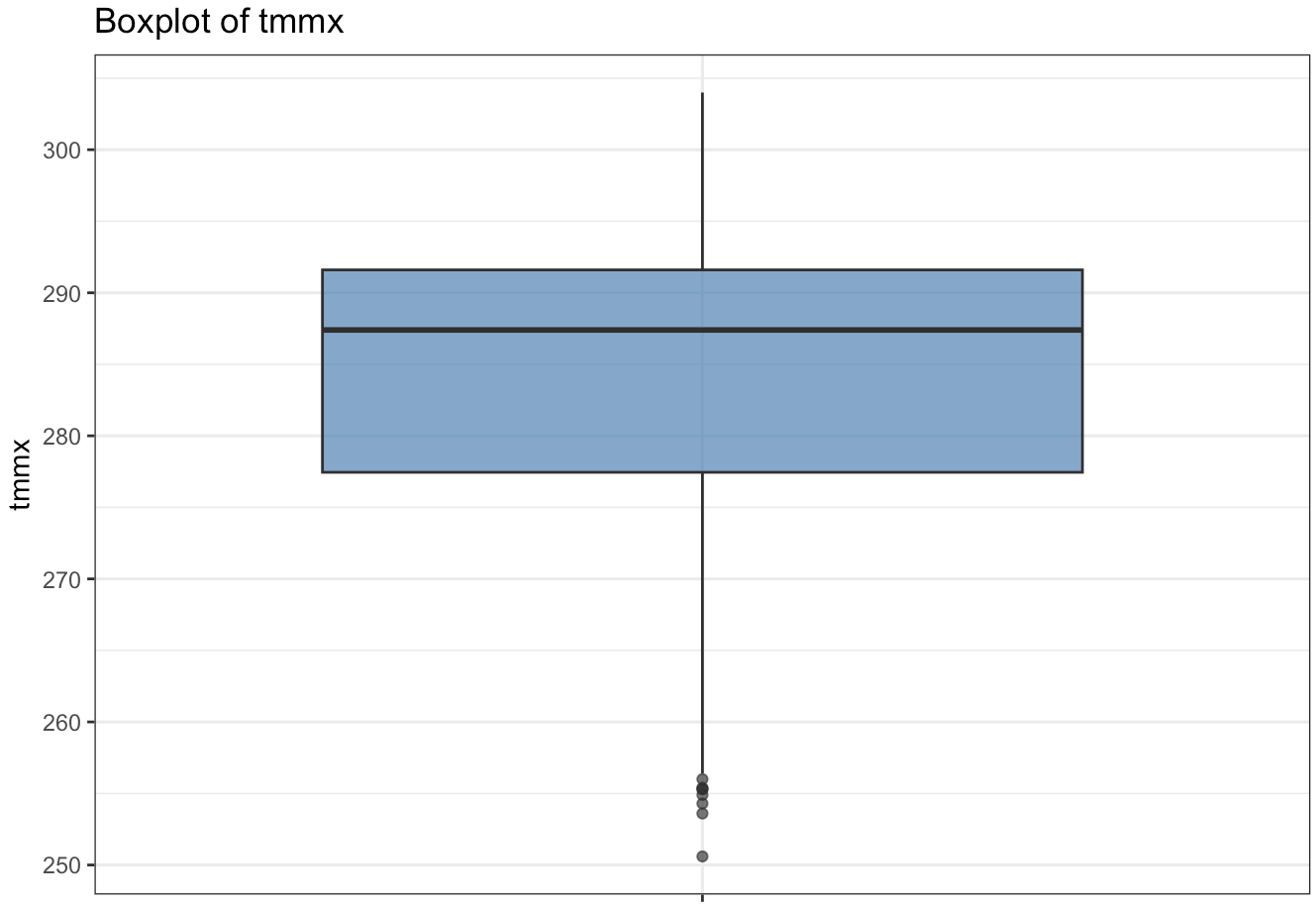

Boxplot of rmax
rmax
100
75
50
25

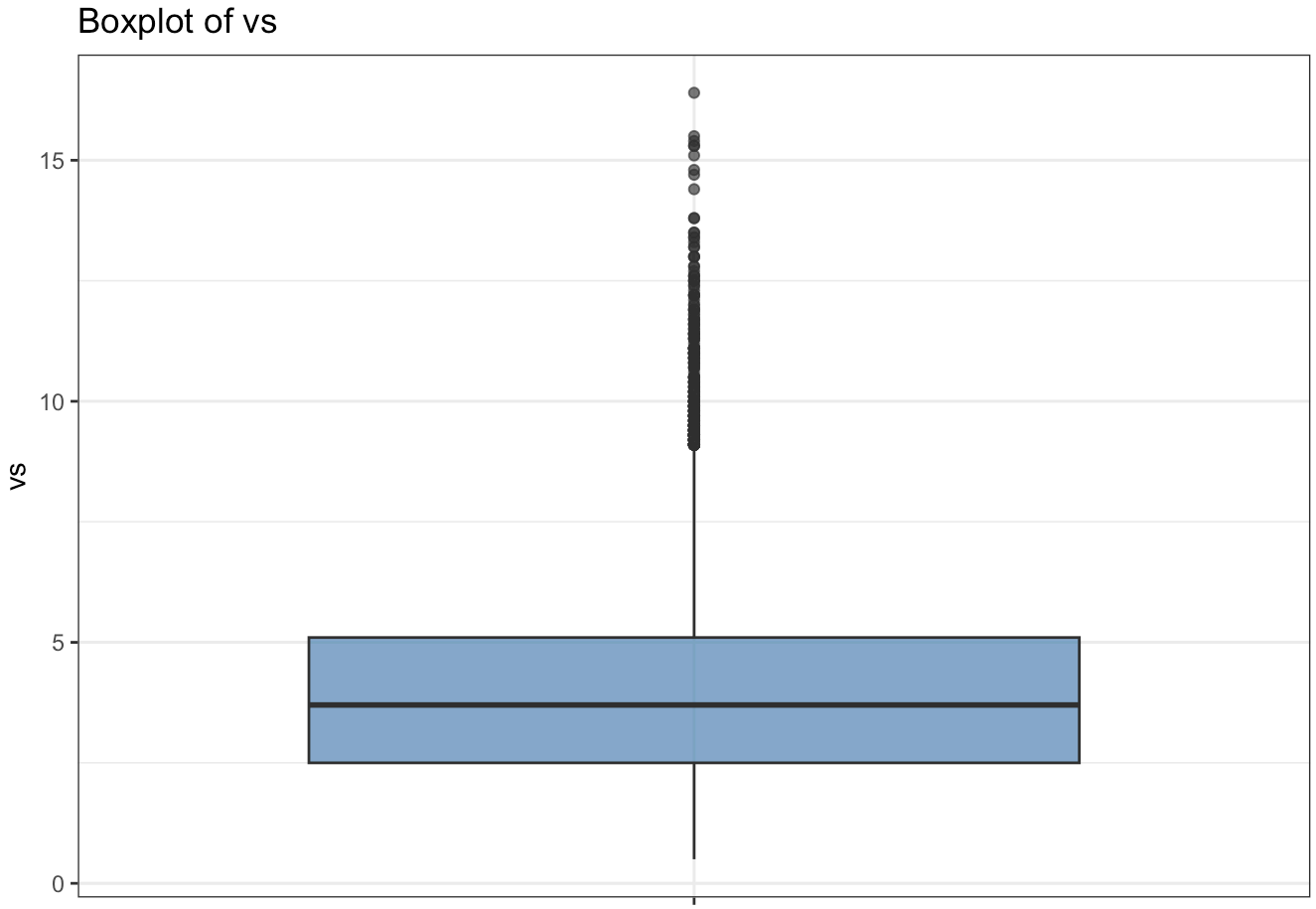

Boxplot of rmin
rmin
100
75
50
25
0

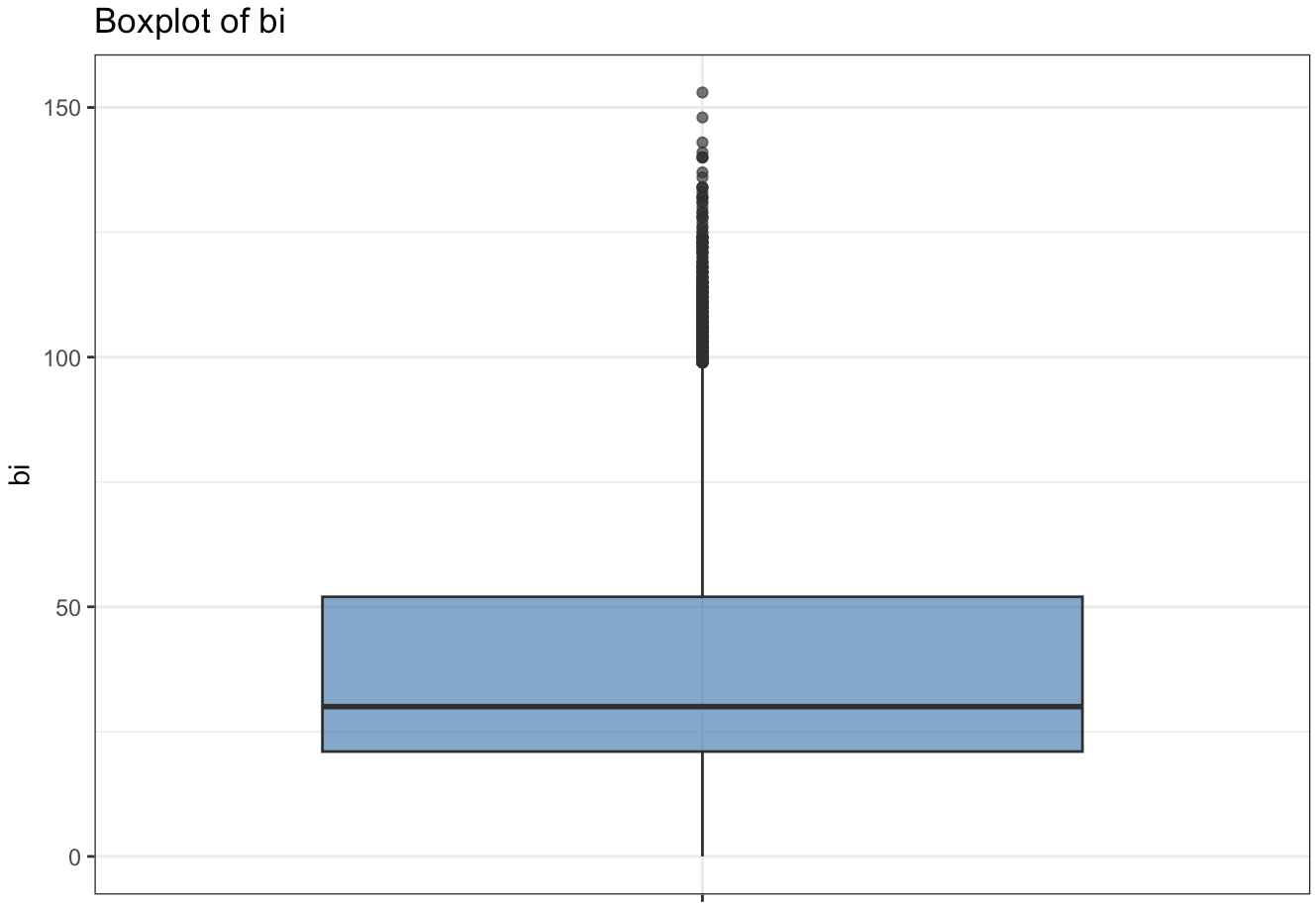

Boxplot of sph
sph
0.010
0.005
0.000

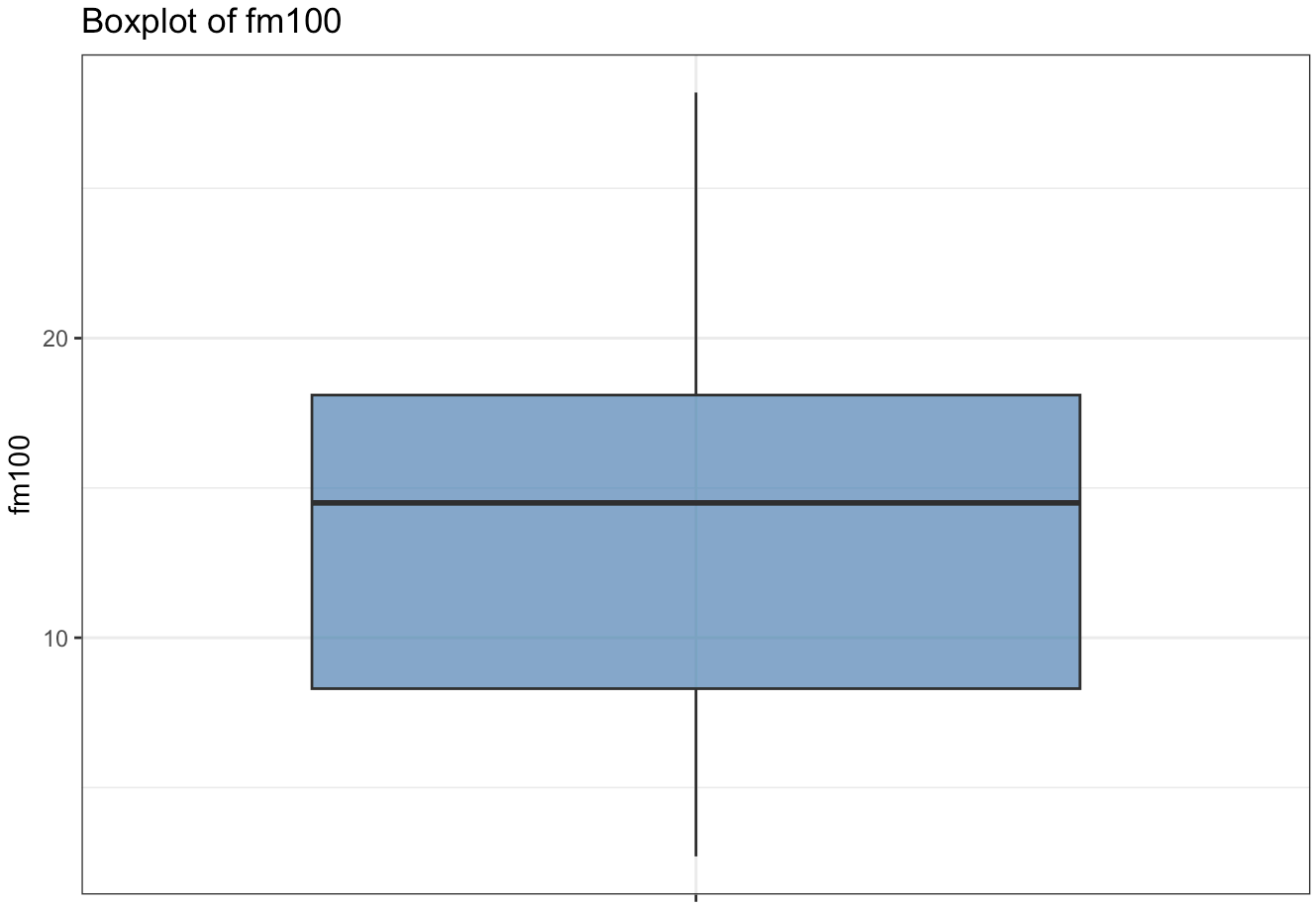

Boxplot of srad
srad
200
100

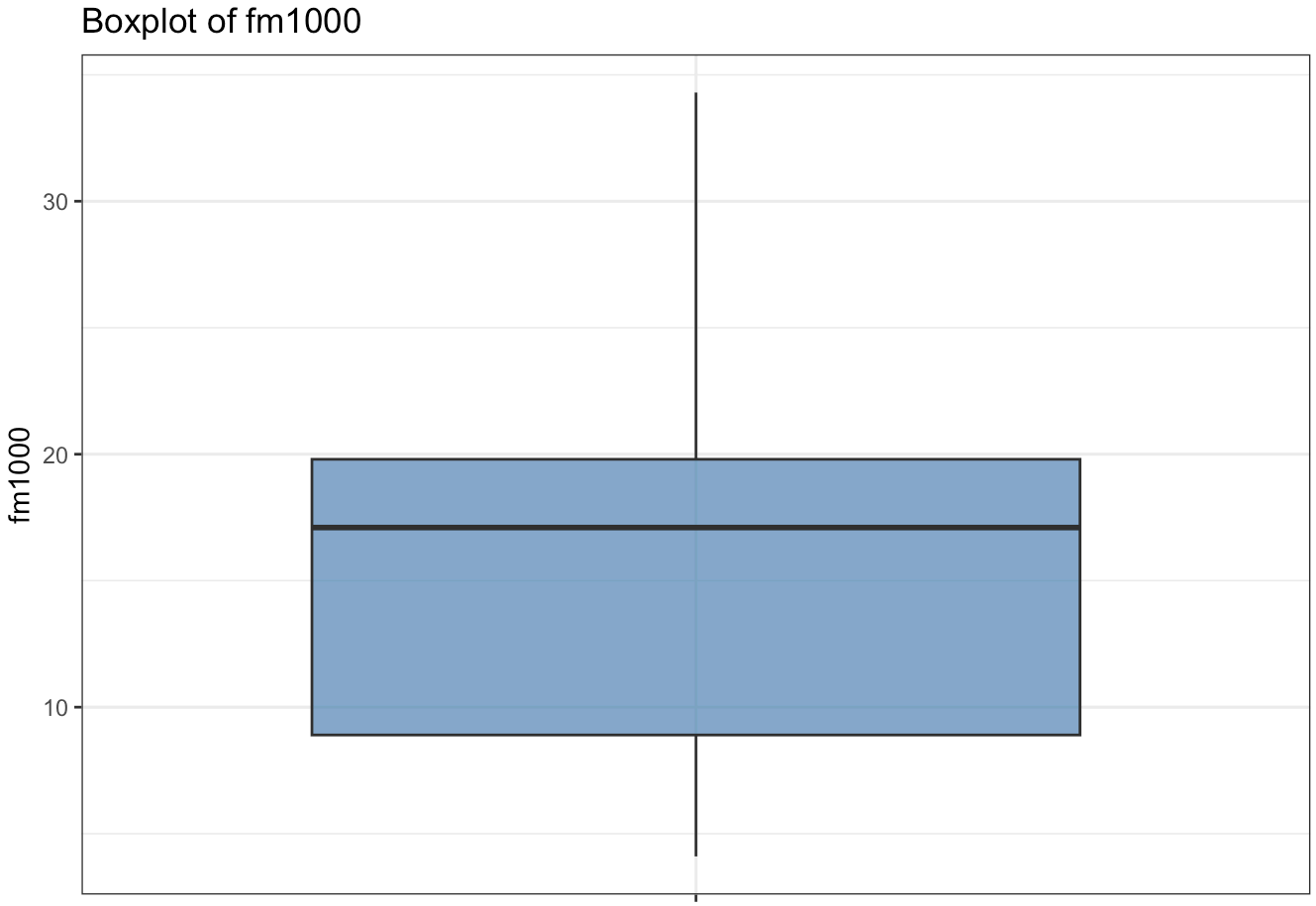

Boxplot of tmmn
300
280
260
tmmn

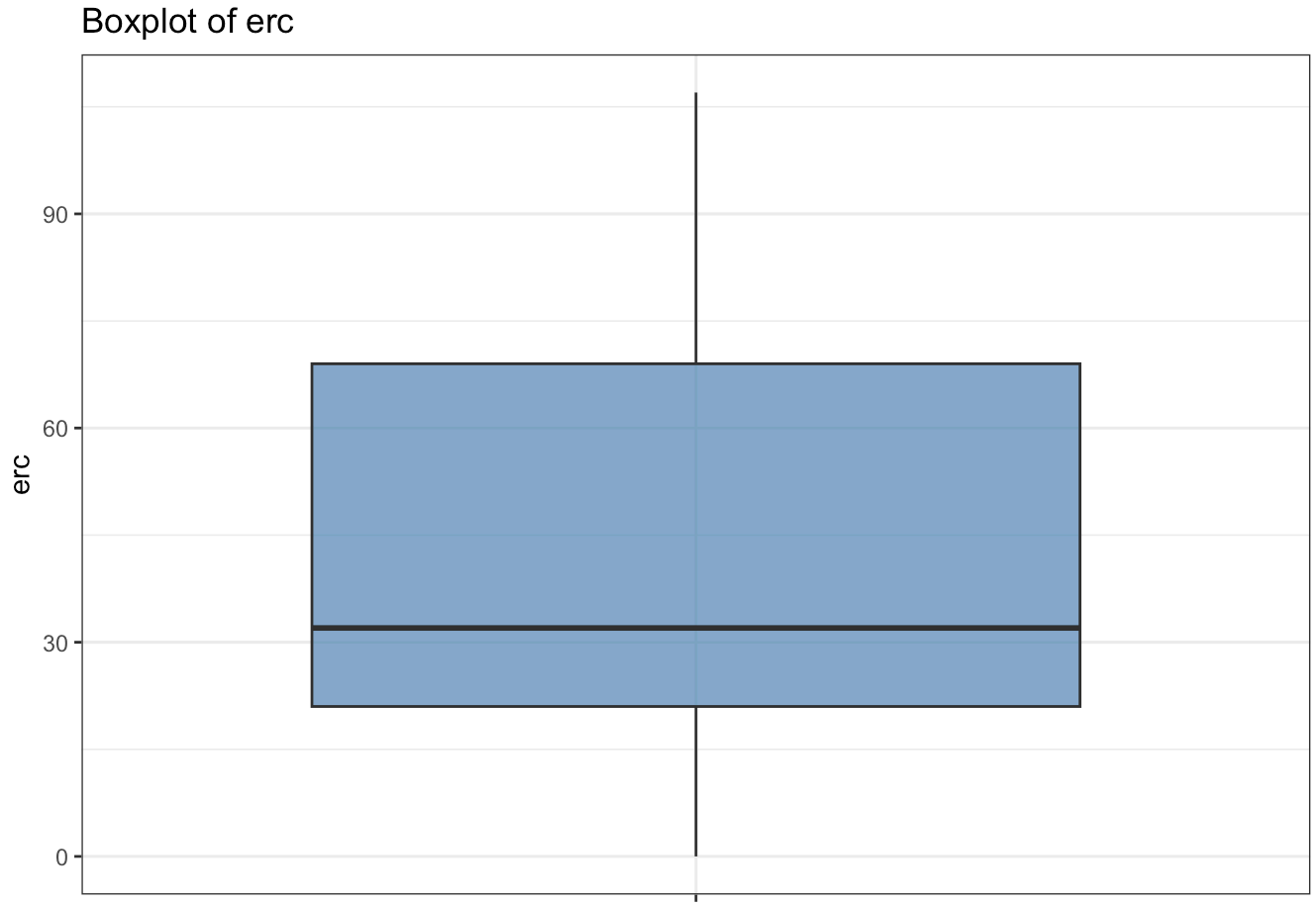

Boxplot of tmmx
tmmx
300
290
280
270
260
250

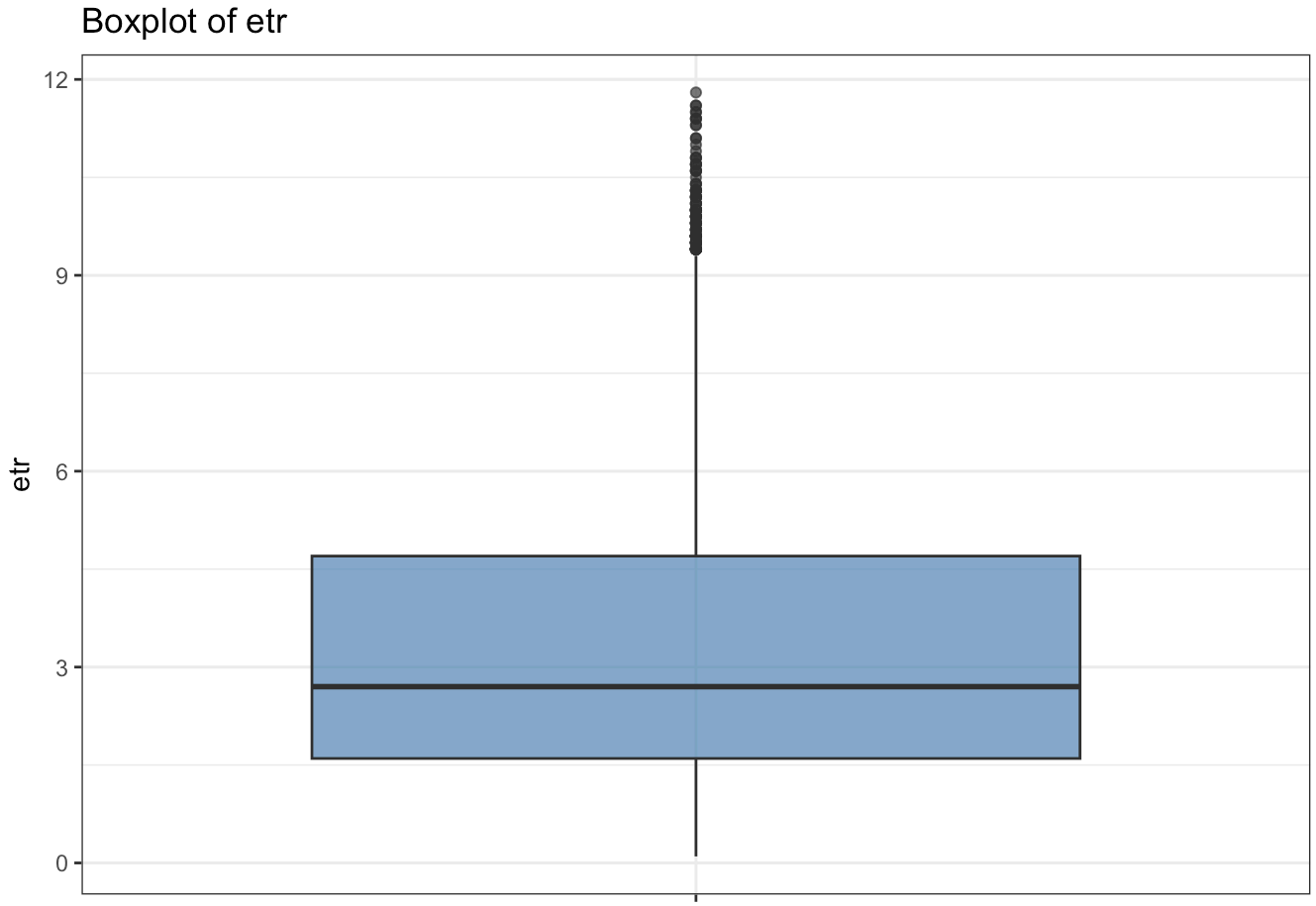

Boxplot of vs
vs
15
10
5
0

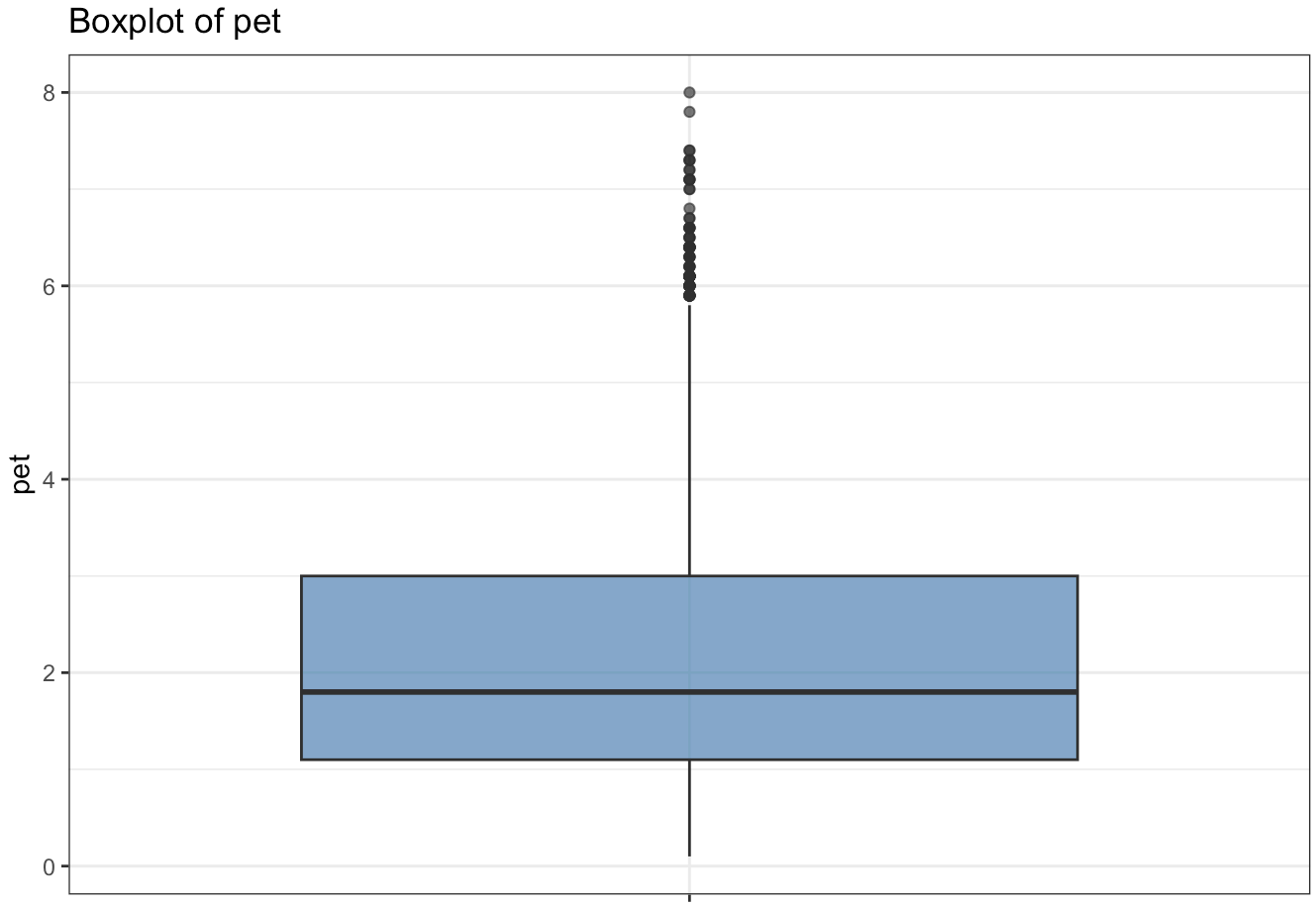

Boxplot of bi
bi
150
100
50
0

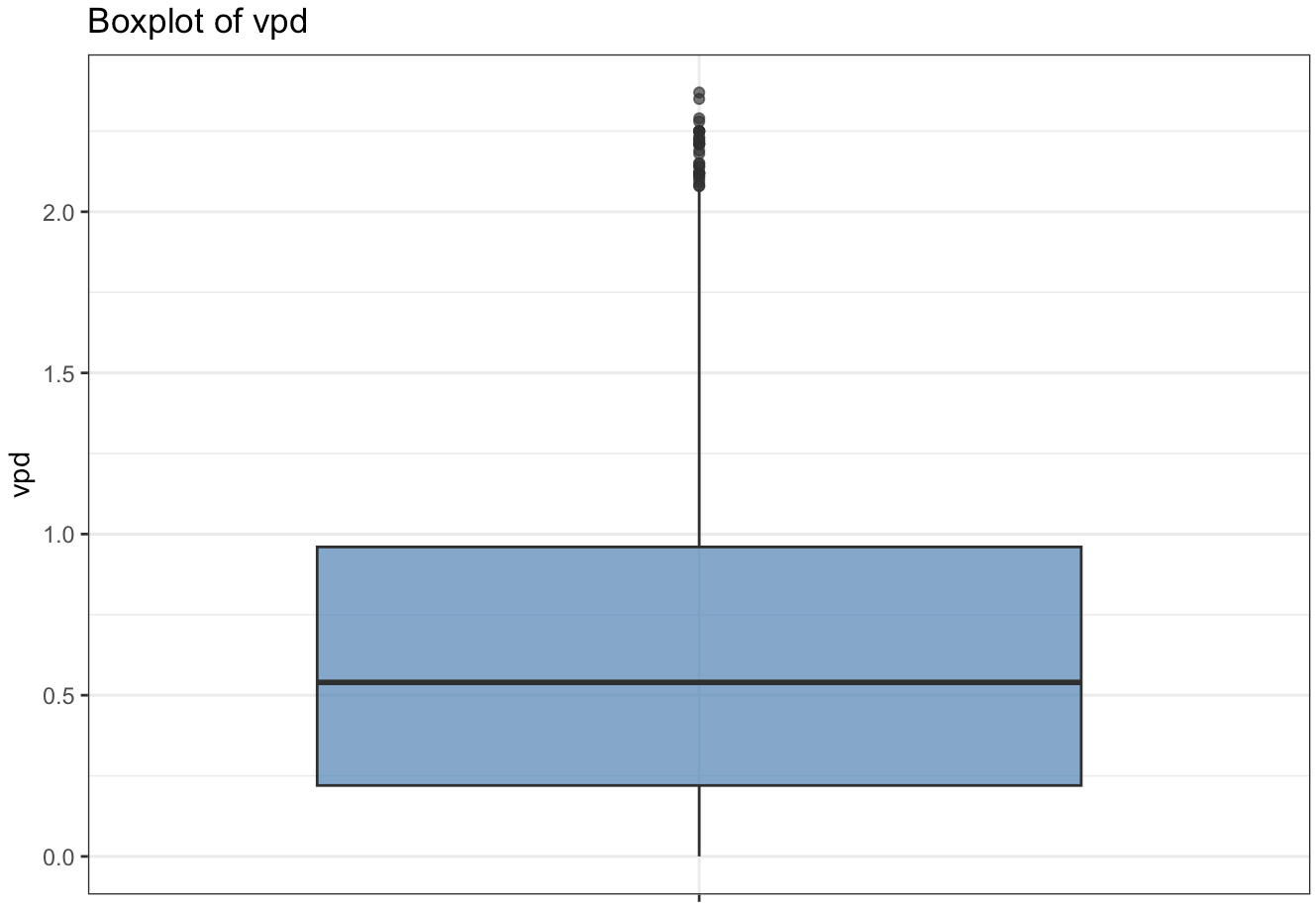

Boxplot of fm100
fm100
20
10


Boxplot of fm1000
fm1000
30
20
10


Boxplot of erc
erc
90
60
30
0


Boxplot of etr
etr
12
9
6
3
0


Boxplot of pet
pet
8
6
4
2
0




```
wildfires_clean <- wildfires_clean %>%
  mutate(region = case_when(
    # northeast: New England + Mid-Atlantic
    latitude >= 37 & latitude <= 47.5 & longitude >= -80 & longitude <= -66.9 ~ "Northeast",

    # south: Southeast + South Central
    latitude >= 24 & latitude < 37 & longitude >= -100 & longitude <= -75 ~ "South",

    # midwest: Great Lakes + Plains
    latitude >= 36.5 & latitude <= 49 & longitude >= -104 & longitude < -80 ~ "Midwest",
```

```
    # southwest: AZ, NM, CO, UT, NV
    latitude >= 31 & latitude <= 42 & longitude >= -125 & longitude < -104 ~ "Southwest",

    # pacific: CA, OR, WA + AK, HI
    latitude >= 32 & latitude <= 49 & longitude >= -125 & longitude < -114 ~ "Pacific",

    TRUE ~ "other"  # catch anything unclassified
  ))
```

```
table(wildfires_clean$region)
```

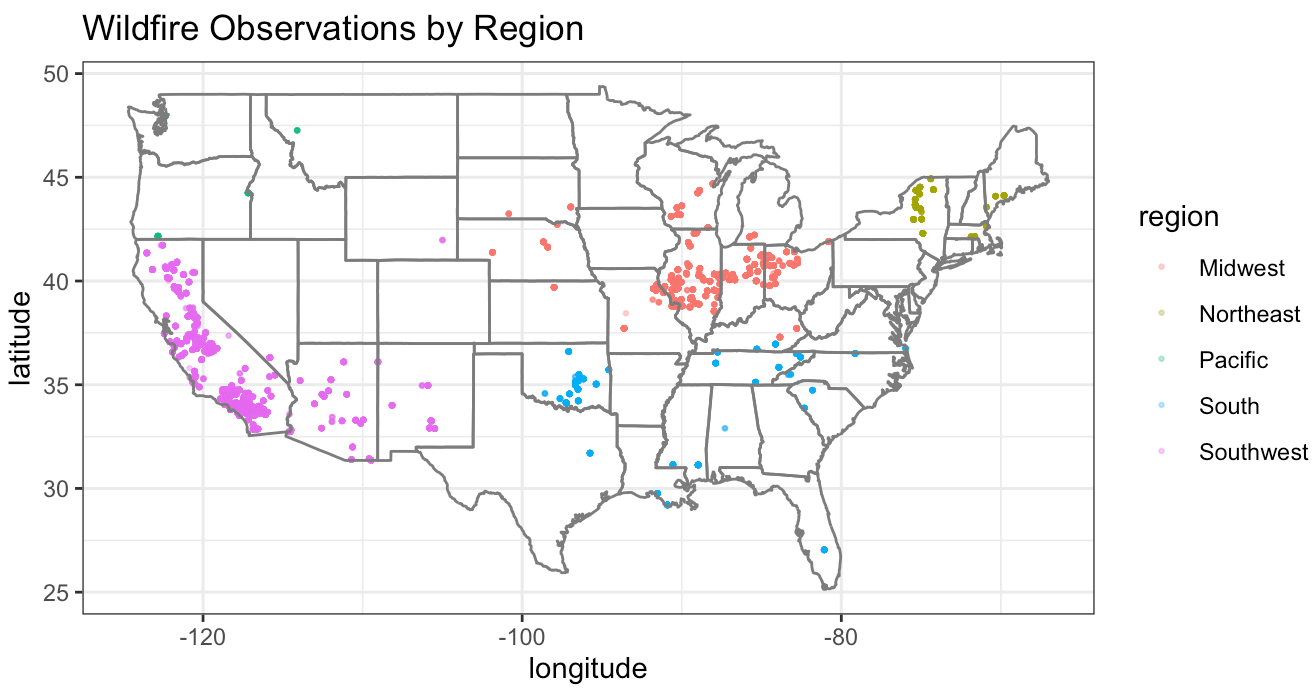
```
 Midwest Northeast   Pacific     South Southwest 
    5575       707       134      1338     10757 
```

```
# check for anything unclassified
wildfires_clean %>% filter(region == "other") %>%
  select(latitude, longitude) %>% summary()
```

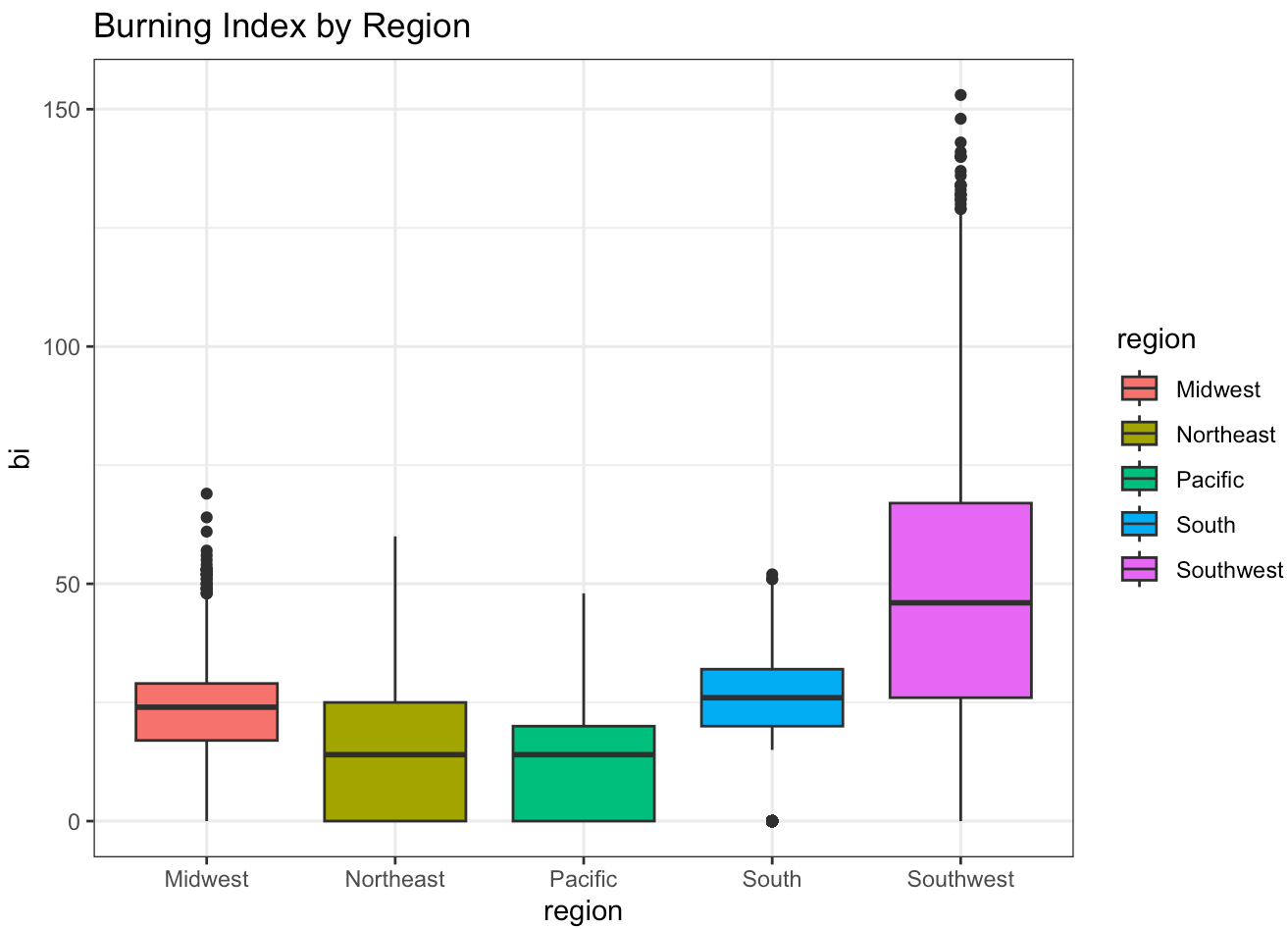
```
    latitude     longitude   
 Min.   : NA   Min.   : NA  
 1st Qu.: NA   1st Qu.: NA  
 Median : NA   Median : NA  
 Mean   :NaN   Mean   :NaN  
 3rd Qu.: NA   3rd Qu.: NA  
 Max.   : NA   Max.   : NA  
```

```
ggplot(wildfires_clean, aes(x = longitude, y = latitude, color = region)) +
  geom_point(alpha = 0.3, size = 0.5) +
  borders("state") +
  coord_fixed(1.3) +
  theme_bw() +
  labs(title = "Wildfire Observations by Region")
```

```
Warning: `borders()` was deprecated in ggplot2 4.0.0.
ℹ Please use `annotation_borders()` instead.
```




```
ggplot(wildfires_clean, aes(x = region, y = bi, fill = region)) +
  geom_boxplot() +
  theme_bw() +
  labs(title = "Burning Index by Region")
```



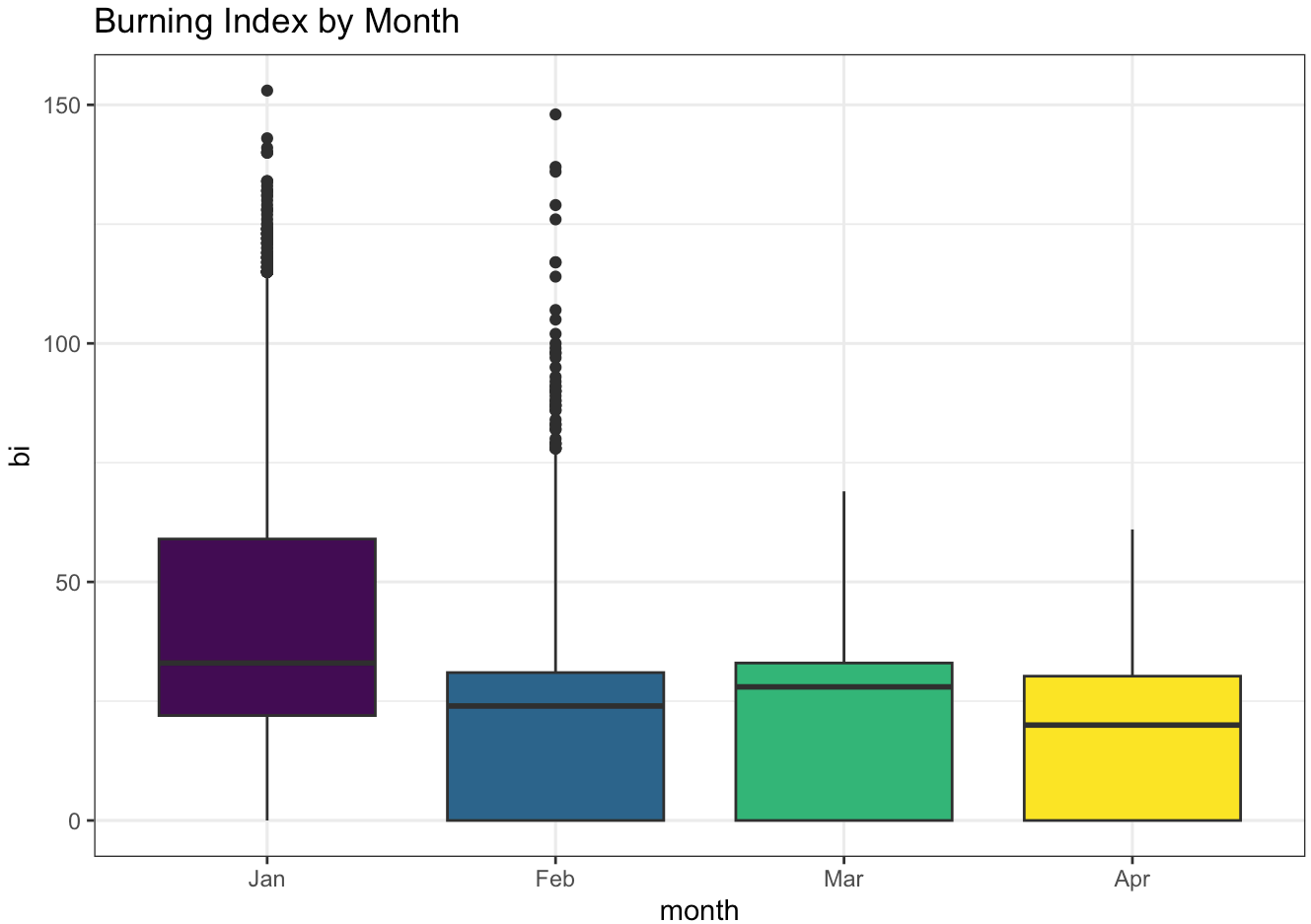

```
wildfires_clean <- wildfires_clean %>%
  mutate(
    Date = as.Date(datetime),   # convert first
    month = month(Date, label = TRUE),
    season = case_when(
      month(Date) %in% c(12, 1, 2)  ~ "Winter",
      month(Date) %in% c(3, 4, 5)   ~ "Spring",
      month(Date) %in% c(6, 7, 8)   ~ "Summer",
      month(Date) %in% c(9, 10, 11) ~ "Fall"
```

```
    )
  )
```

```
head(wildfires_clean %>% select(Date, month, season))
```

```
# A tibble: 6 × 3
  Date       month season
  <date>     <ord> <chr>
1 2025-01-01 Jan   Winter
2 2025-01-02 Jan   Winter
3 2025-01-03 Jan   Winter
4 2025-01-04 Jan   Winter
5 2025-01-05 Jan   Winter
6 2025-01-06 Jan   Winter
```

```
ggplot(wildfires_clean, aes(x = month, y = bi, fill = month)) +
  geom_boxplot() +
  theme_bw() +
  theme(legend.position = "none") +
  labs(title = "Burning Index by Month")
```

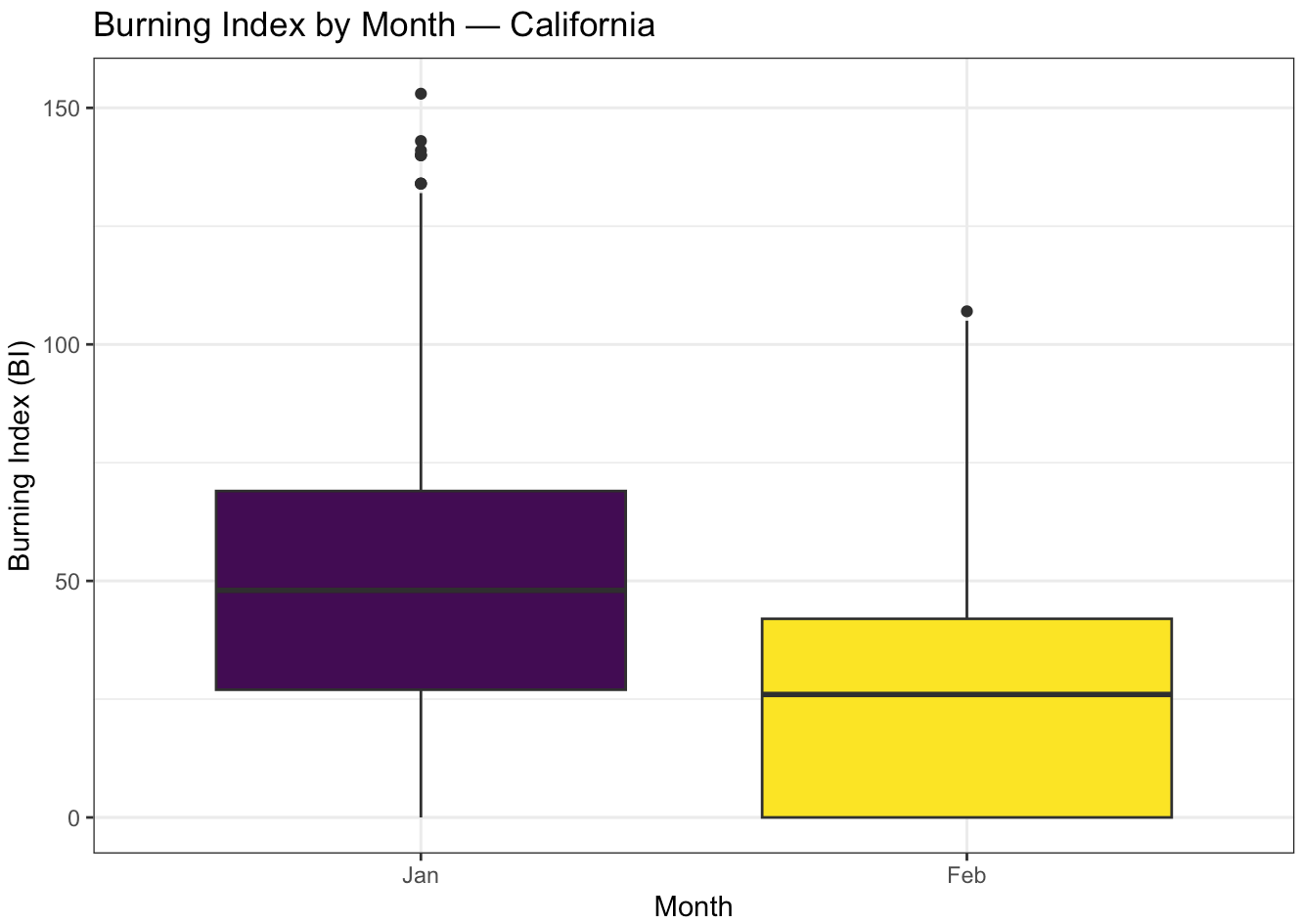



```
california <- wildfires_clean %>%
  filter(latitude >= 32.5 & latitude <= 42.0,
         longitude >= -124.5 & longitude <= -114.0)

nrow(california)  # rows
```

```
[1] 10038
```

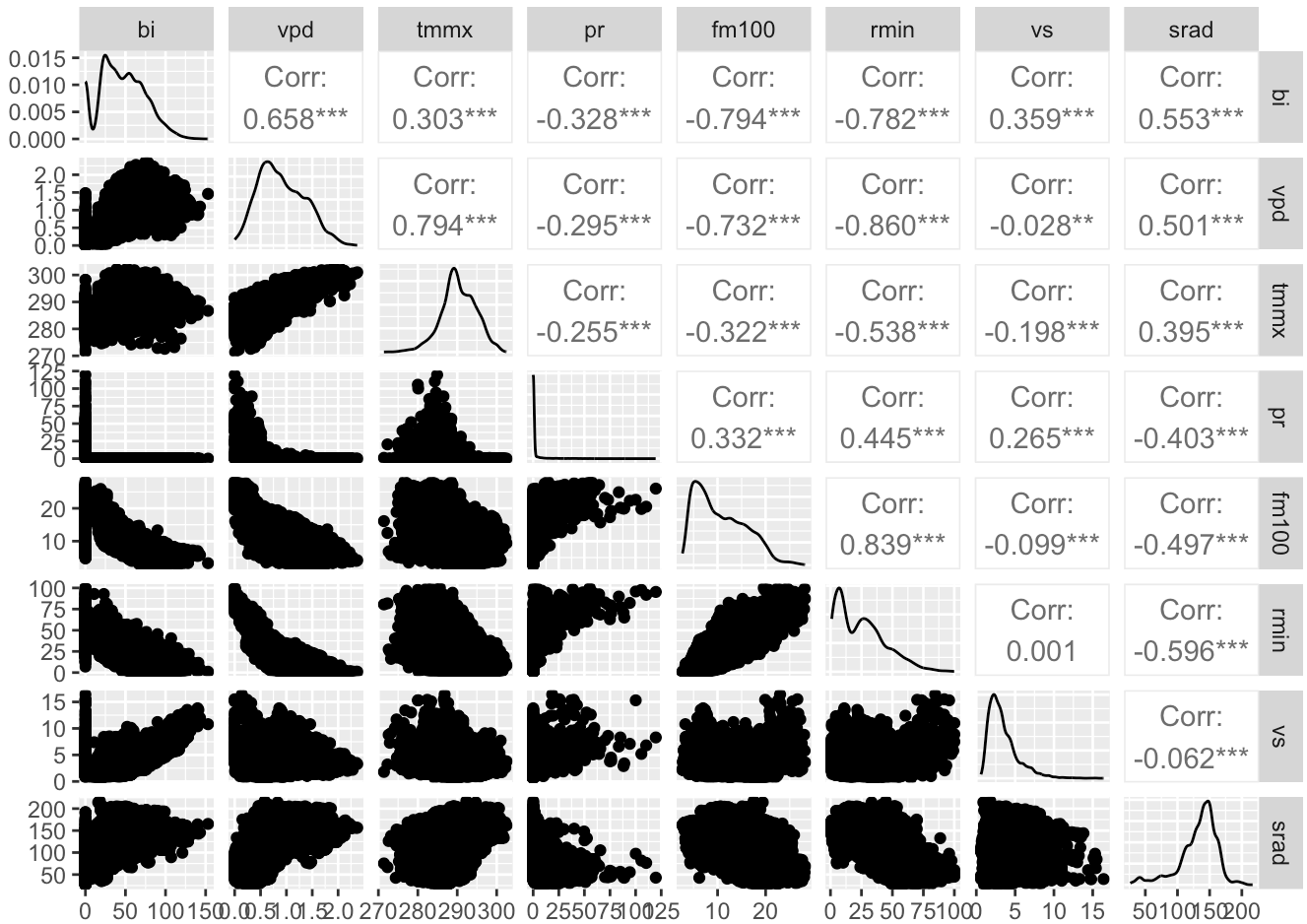
```
ggplot(california, aes(x = month, y = bi, fill = month)) +
  geom_boxplot() +
```

```
  theme_bw() +
  theme(legend.position = "none") +
  labs(title = "Burning Index by Month — California",
       x = "Month", y = "Burning Index (BI)")
```




```
california %>% count(month)
```

```
# A tibble: 2 × 2
  month     n
  <ord> <int>
```

```
1 Jan     8679
2 Feb     1359
```

```
california %>%
  select(bi, vpd, tmmx, pr, fm100, rmin, vs, srad) %>%
  ggpairs()
```